UMRS Restoration Features

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Preface

This restoration features book documents the key design elements of ecosystem restoration features built on the Upper Mississippi River System (UMRS). This data model has been developed to represent the wide range of restoration practices employed on the UMRS throughout many years and be adaptable into the future. As we learn lessons about restoration, this data model has evolved along with our best practice.

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Abstract

1 Introduction

1.1 What are Restoration Features?

Restoration features are what we build.

1.2 UMRS Restoration Data Model

Purpose

The restoration features are GIS feature classes designed to document the key design elements of ecosystem restoration features built on the Upper Mississippi River System (UMRS). This data model has been developed to represent the wide range of restoration practices employed on the UMRS throughout many years and be adaptable into the future. As we learn lessons about restoration, this data model has evolved along with our best practice.

Data Design Standards

The entities in this data model represent the physical structures constructed to accomplish the restoration objectives of each project. These entities are grouped into standardized GIS "feature classes" (groups of objects of single type having a common set of attributes). This standard was adapted from the DOD SDSFIE-V (https://www.sdsfie.mil) to represent the specialized features built on the UMRS. This data model was designed to represent UMRS best practice as documented in the UMRR Design Handbook (2012, 2025).

Data Use

- Common Terms - Provides a common lexicon of ecosystem restoration features representing current best practice to streamline communication across a diverse group of stakeholders, spread across many organizations, from many disciplinary backgrounds.
- Cartography - Defines a standardized framework for representing a wide variety of complex spatial features on different media types (i.e., static paper maps, interactive web maps). Maps are one of the most effective communication tools to describe project benefits to the public.

- Upward Reporting - Expressing these diverse features in a standard framework supports automated summarization to streamline routine upward reporting of project status and benefits.
- Record Keeping - Forty years of UMRS restoration has generated a rich set of built features to monitor and learn from. This standardized framework, that can evolve with current best practice, is the best way to track our investments.
- Whole Project Lifecycle - This standardized framework allows us to track projects throughout their entire lifecycle, from planning and feasibility, through design and construction, and through operation and maintenance. UMRS Restoration Data Model

1.3 Implementation

The Army Corps of Engineers project process typically involves four main phases: pre-study initiation (community request), feasibility study, design and construction, and operation and maintenance.

Feasibility

- The Corps conducts a detailed study to assess the problem, evaluate potential solutions, and determine project feasibility.
- Analysis includes environmental impacts, economic benefits, and cost-benefit analysis.
- Report is submitted to higher authorities for review and approval.

Design & Construction

- If the project is approved, detailed design plans are developed.
- Necessary permits are obtained.
- Construction contracts are awarded and project is built.

Operation and Maintenance

Once completed, the Corps manages the project, including routine maintenance and repairs.

1.4 Current Programs Usage

UMRR

To work within a partnership among federal and state agencies and other organizations; to construct high-performing habitat restoration, rehabilitation projects; to produce state-of-the-art knowledge through monitoring, research, and assessment; to engage other organizations to accomplish the Upper Mississippi River Restoration Program's vision.

The HREP element has fostered interdisciplinary and collaborative planning for habitat restoration, preservation, and enhancement previously non-existent on any other river system in the United States.

NESP

The Navigation and Ecosystem Sustainability Program (NESP) is a long-term, dual-purpose program that integrates navigation improvements and ecosystem restoration together to provide Upper Mississippi River System once in a generation-type positive impact.

The purpose of the Navigation and Ecosystem Sustainability Program is to improve efficiency and capacity of the nationally significant Upper Mississippi River – Illinois Waterway (UMR-IWW) navigation system while protecting preserving and enhancing the structure, diversity, and function of this nationally significant ecosystem.

Notes

Several ecosystem restoration programs are active on the UMRS at the local, state, and federal levels. This data model was designed to function as a repository of the ecosystem restoration features common within a large river floodplain system such as the UMRS, independent of jurisdiction or funding source.

2 Case Studies

An Integrated Framework in Practice: Detailed Case Studies in USACE Upper Mississippi River Restoration

Introduction: From a Toolkit of Features to Integrated Systems

The ecological restoration of the Upper Mississippi River System (UMRS) is a monumental undertaking guided by the sophisticated methodology outlined in the U.S. Army Corps of Engineers' (USACE) *2012 UMRR EMP Environmental Design Handbook*. While the handbook provides a detailed lexicon of individual "restoration features," their true power is only realized when they are integrated into a cohesive project designed to address specific, large-scale ecological deficits. To understand this methodology in its practical application, it is best to examine it through the lens of real-world case studies. This chapter will therefore explore the USACE restoration process through four major Habitat Rehabilitation and Enhancement Projects (HREPs). Each section will deconstruct a project to show how a unique combination of primary and supporting features was used to address its specific ecological problems, demonstrating how the abstract toolkit is transformed into functional, resilient, and interconnected ecosystems on the ground.

Case Study 1: The Pool 8 Islands HREP - Geomorphic Restoration for Ecological Resurgence

Project Context and Goal: The Pool 8 Islands HREP, located in the vast Weaver Bottoms area near Brownsville, Minnesota, addressed one of the most significant consequences of river impoundment: the creation of a wide, shallow, lake-like environment. The 4,000-acre backwater suffered from a wind fetch of over two miles, which generated constant wave action. This energy re-suspended fine bottom sediments, leading to extremely high turbidity that blocked sunlight, prevented the growth of vital submerged aquatic vegetation (SAV), and eroded the few remaining natural shorelines. The primary goal was to re-engineer the geomorphology of the pool to reduce wave energy, improve water clarity, and spur the large-scale recovery of the aquatic ecosystem.

Integrated Feature Application in Detail: This project is a classic example of geomorphic restoration centered on the beneficial use of dredged material. The foundational activity was a large-scale **Dredge Event**, which sourced over 1.5 million cubic yards of clean sand during routine maintenance of the 9-foot navigation channel. This material was then transported and strategically placed to construct a complex of 17 new **Islands**. The curvilinear, naturalistic layout of these islands was precisely guided by a detailed **Island Line** survey plan, which provided the exact coordinates for construction crews to place material accurately and create the desired footprint.

The immediate stability of these new sand landforms was critical. To prevent their immediate erosion by the very forces they were designed to block, their exposed, windward shores were armored with a heavy rock **Revetment**. This involved placing thousands of tons of Class III riprap over a durable geotextile fabric, creating a resilient shield against wave attack that was essential for the project's long-term survival.

Once the physical forms were stable, a massive **Flora Planting or Seeding** effort transformed the sterile sand into living habitat. On the highest elevations of the islands, designed to be above the 2-year flood frequency, over 80,000 hardwood trees—including species like swamp white oak, river birch, and pecan—were planted to initiate a floodplain forest. On the lower, marshy benches and shorelines, the substrate was seeded with native grasses and forbs to create a diverse emergent **Wetland** fringe, providing both erosion control and valuable habitat.

The project's success was profound. In the protected zones behind the islands, water clarity improved from inches to several feet. This triggered a massive resurgence of SAV, with coverage increasing by hundreds of acres. This, in turn, created a rich nursery and foraging habitat for fish and invertebrates, and a major feeding area for migratory waterfowl. To manage the area and inform the public, numerous **Signs** were installed. These include kiosks at public access points explaining the project's success and "Slow No Wake Zone" buoys to protect the fragile, newly established vegetation from boat wakes.

Case Study 2: The Weaver Bottoms (Pool 5) HREP - Large-Scale Drawdown and Geomorphic Control

Project Context and Goal: The Weaver Bottoms project in Pool 5 was another massive HREP aimed at restoring a 4,000-acre backwater that had degraded into a wide, shallow lake with almost no vegetation. Like Pool 8, it suffered from immense wind and wave action. The project had a dual goal: first, to reduce wave energy through physical barriers, and second, to stimulate the germination of the dormant but viable native seed bank that lay buried in the riverbed sediments, a process that required physically dewatering a vast area of the Mississippi River.

Integrated Feature Application in Detail: This complex project combined geomorphic construction with large-scale hydraulic manipulation. Similar to Pool 8, a major **Dredge Event** sourced material for the construction of several new **Islands**, including a large, crescent-shaped island, to break up wave fetch and create sheltered zones. However, the most innovative component was a large-scale, partial **drawdown** of the backwater. This was a significant engineering challenge that required modifying existing **Water Control Structures** at the outlet of the backwater to release water faster than it was entering. To facilitate the dewatering of such a vast area, targeted **Dredging** was also performed to create deeper **Flow Channels** that would act as drainage arteries, ensuring the entire area could dewater effectively as the main pool level was lowered.

The drawdown exposed thousands of acres of mudflats for an entire growing season. This had two key ecological effects: first, it allowed the wet, mucky sediment to dewater and consolidate, creating a firmer substrate. Second, and more importantly, the combination of sunlight and air stimulated the germination of a massive seed bank of native aquatic plants (like wild celery and arrowhead) that had been dormant in the sediment for years. When the area was slowly reflooded the following year, a lush and diverse plant community emerged, transforming the "dead lake" back into a productive marsh. The newly constructed islands, protected by **Revetments**, provided the long-term wave protection necessary to ensure this newly established vegetation would persist.

Case Study 3: The Swan Lake HREP - Active Management through Complex Infrastructure

Project Context and Goal: The Swan Lake HREP, a vast 10,000-acre floodplain complex on the Illinois River, is one of the most ambitious examples of active habitat management. The site had become disconnected from natural flood pulses, and its wetlands were degraded. The primary goal was to restore a highly productive mosaic of wetland habitats by giving resource managers the ability to control water levels with absolute precision, completely independent of the unpredictable and often ill-timed fluctuations of the river.

Integrated Feature Application in Detail: This project is a masterclass in the integration of supporting infrastructure to achieve an ecological goal. The project's core is a series of large **Moist Soil Management Units (MSUs)**. This first required the construction of miles of new **Embankments**, whose alignment was precisely dictated by an **Embankment Centerline**. The design was a major geotechnical challenge, requiring extensive soil analysis from data collected at dozens of **Borehole Points** to ensure the embankments would be stable on the soft, alluvial floodplain soils.

The ability to manage water is enabled by several key features. Large, cast-in-place concrete **Water Control Structures**, fitted with removable stoplogs, were built into the embankments.

Because gravity flow was insufficient, several large **Pump Stations** were constructed, with a combined capacity of over 300 cubic feet per second. This immense power gives managers the ability to fill the vast wetland units from the river in the fall or, critically, to dewater them after an unseasonal summer flood to save the growing food crop.

This operational infrastructure required an equally complex network of utilities and access. A robust electrical grid was installed, consisting of miles of buried armored cable (**Electrical Utility Segments**) that run along the levee crests. These segments connect the main power grid to the **Electrical Utility Nodes**—the transformers and motor control panels—located at each remote pump station. A water conveyance network, composed of large-diameter HDPE pipes (**Water Utility Segments**) and large, wheel-operated gate valves (**Water Utility Nodes**), moves the massive volumes of water from the pumps to the various wetland cells.

None of this would be possible to build, operate, or maintain without physical access. The project included the construction of several miles of **Road Segments** on the crests of the main embankments. These roads were surfaced with a thick **Pavement Section** of compacted aggregate, specifically designed to support the weight of heavy vehicles like cranes and excavators needed for long-term maintenance. The entire construction process was guided by a **Feature Stationing** system, which provided a linear "address" for every component, allowing for the precise placement of every culvert, pump, and water control structure along the project's miles of embankments.

Case Study 4: Restoring Connectivity and Complexity - Peoria Lake, Pool 12, and Big Timber HREPs

Project Context and Goal: Many HREPs are not about large-scale geomorphic change but instead focus on more surgical interventions to restore depth, flow, and habitat complexity in degraded backwaters and side channels that have become choked with sediment and cut off from the main river.

Integrated Feature Application in Detail: At the **Peoria Lake HREP**, the foundational activity was a **Dredge Event** that removed over 1.3 million cubic yards of sediment to excavate deep channels in the sediment-filled lake. Because the dredged material was fine-grained silt, it was unsuitable for island building and was instead pumped to a designated **Dredge Placement** site—an adjacent upland containment area where it could be safely dewatered. To bring the newly dredged pools to life, a new **Flow Channel** was excavated to connect them to the main river, providing a constant supply of fresh water. Within these pools, **Fish Structures**, including rock piles and anchored logs, were placed to provide immediate habitat complexity for the local fishery.

In **Pool 12** of the Mississippi River, the focus was on restoring flow to side channels cut off by historic **Riverine Flow Structures** (rock wing dikes). Rather than removing the dikes,

a more targeted approach was taken. A **Notch**, typically 30-50 feet wide, was cut into each dike. The design and placement of these notches were guided by a **Riverine Flow Structure Line** survey plan to ensure they captured the right amount of flow. This allowed a controlled portion of the river's current to pass through, scouring the side channel and restoring a healthy, flowing environment without compromising the navigation function of the main channel.

Finally, at the **Big Timber HREP** in Pool 17, managers faced a different problem: the water in the adjacent Mississippi River channel was often too turbid and nutrient-laden to be desirable for their managed wetland. To overcome this, the project included the installation of a high-capacity **Well**. This feature drills deep into the alluvial aquifer to provide a source of clean, clear, sediment-free groundwater. The well gives managers a critical tool to fill the wetland with high-quality water, allowing them to achieve their vegetation goals even when the river itself is not a suitable water source.

3 Restoration Features Summary

The Integrated Design of River Restoration Features

The "2012 UMRR EMP Environmental Design Handbook" outlines a comprehensive and integrated approach to river restoration, where a diverse suite of features are designed to work together to achieve large-scale ecological uplift in the Upper Mississippi River System. The relationship between the individual restoration features and the overarching design practices is not linear but rather a complex, synergistic web. The core design practice is to view the river not as a series of isolated problems but as a complete ecosystem. Therefore, projects combine multiple features to mimic natural processes, restore habitat, and ensure the long-term resilience of the restored environment.

The restoration features can be broadly understood in two categories: Primary Ecological Features and the Supporting Engineering and Infrastructure Features that enable them. The design practice involves layering these two categories to create a functional and sustainable project.

1. Primary Ecological Features: The Building Blocks of Restoration These features are the visible, direct interventions aimed at restoring ecological structure and function. Design practices for these features focus on creating specific habitats and restoring natural river processes.

- Objective: Enhancing Aquatic Habitat & Connectivity
  - Features Used: Dredge Events are a cornerstone practice to restore depth in sediment-filled backwaters. This creates overwintering refugia for fish and improves flow. The design is not simply about digging a hole; it involves creating varied depths and complex shapes. To further enhance this, Fish Structures (like rock piles and log snags) are added to provide cover. Flow Channels and Notches in dikes are engineered to reintroduce vital, oxygenated water from the main channel into these restored backwaters, preventing stagnation.
  - Relationship to Design: The design practice here is one of habitat layering. A single backwater dredging project is designed in concert with flow introduction and in-stream cover to create a complete, multi-faceted aquatic habitat, rather than just a deep pool.
- Objective: Restoring Landforms and Floodplain Dynamics

  - Features Used: Islands, often built using material from Dredge Events (a key synergy), are a primary tool. Their design is strategic: they act as barriers to reduce wind and wave erosion in backwaters, creating sheltered zones for aquatic plants to grow. The islands themselves are designed with varied topography, using Flora Planting and Seeding to establish Floodplain Forests and provide secure nesting habitat.
  - Relationship to Design: The design practice is geomorphic restoration. It's not just about building an island, but about using that island as a tool to reshape the river's flow, energy, and light conditions to benefit the wider ecosystem. The island's design (shape, elevation, vegetation) is dictated by these broader goals.

- Objective: Managing Riparian and Wetland Zones
  - Features Used: Moist Soil Management Units (MSUs) and Wetlands are created or restored using Embankments to impound water. The key design practice is the precise management of water levels, which is accomplished using Water Control Structures, Pump Stations, and Wells. By controlling the timing and duration of flooding and drawdowns, managers can stimulate the growth of specific native plants that provide critical food for migratory waterfowl.
  - Relationship to Design: This demonstrates the practice of active ecological management. The features are designed not as static elements, but as a dynamic, controllable system that allows managers to adapt to environmental conditions and meet specific ecological targets year after year.

2. Supporting Engineering and Infrastructure Features: The Foundation of Success These features are the critical engineering components that make the primary ecological restorations possible, stable, and accessible. The design practice is to integrate these robust engineering solutions seamlessly to ensure the project's longevity and functionality.

- Geotechnical and Structural Stability: A Revetment (rock armor) is designed to protect a newly built Island from eroding. The design of a massive Embankment relies entirely on data from Borehole Points to understand the soil's capacity to support the structure. Riverine Flow Structures like wing dikes are designed based on extensive hydraulic modeling to ensure they steer river currents effectively without causing unintended erosion elsewhere.
- Construction and Operational Logistics: The entire system is built upon a framework of civil infrastructure. Feature Stationing, Embankment Centerlines, and Island Lines provide the precise survey control needed for construction. Road Segments and Pavement Sections are designed to provide all-weather access for heavy machinery during construction and for long-term maintenance and monitoring.

- Power and Utilities: The active management of an MSU is impossible without a reliable Pump Station, which in turn depends on a network of Electrical Utility Nodes and Segments. The design practice here ensures that this critical infrastructure is itself resilient, often elevated above flood levels to maintain operational capacity during high-water events.

Conclusion

In summary, the relationship between restoration features and design practices is one of interdependence and synergy. The design process for a UMRR EMP project begins with a clear ecological objective, such as improving fish habitat or restoring a floodplain forest. To achieve this, a suite of primary ecological features are selected and designed as an interconnected system. Finally, this system is built upon a robust foundation of supporting engineering and infrastructure features, which are designed to ensure the primary features are constructed correctly, remain stable over their design life, and can be actively managed to maximize their ecological benefits. No single feature stands alone; its design is always considered in the context of the entire project and the broader river ecosystem.

Part I

Restoration Features

4 Restoration Features

4.1 Purpose

Restoration of ecosystems through a variety of features, including wetlands, riparian areas, aquatic systems, and oyster reefs. The goal of these restoration projects is to restore the natural function, structure, and processes of degraded ecosystems.

4.2 Notes

Data Sources - If a project has been constructed, CAD drawings or As-Built drawings should be used to guide feature digitizing. If a project is in the planning stages, the schema can be used to record potential features and track which planning alternatives they participate in. Elevation data should be attached to geometries as specified in this document.

These feature classes are stored in the MVR UMRR Oracle database to ensure that authoritative data is available to all. See these reference guides to access the oracle database for editing features.

Restoration features are a topical group of feature classes used to represent, at various phases, engineering features constructed or implemented to directly meet the goals and objectives of programs and the projects they support.

4.3 Restoration Features

- DredgingEvent (polygon) – The extent of the area subject to dredging action. (area of material removal)
- Embankment (polygon) – The centerline of an embankment for controlling the waters of the sea, river, or other water bodies.
- EmbankmentCenterline (line) – The centerline of an embankment for controlling the waters of the sea, river, or other water bodies.
- FishStructure (polygon) – An engineered structure designed to improve fish habitat or allow improved fish movement through the river system.
- FloraPlantingOrSeeding (polygon) – A location where vegetation is planted or seeded.

- FlowChannel (polygon) – An excavated channel (via mechanical or hydraulic means) intended to allow for the flow of water across, into, or out of the project area or any subcomponent of a project area.
- Island (polygon) – Island structures engineered with a variety of techniques and materials that serve as barriers to flow, wind and waves.
- IslandLine (line) – Island structures engineered with a variety of techniques and materials that serve as barriers to flow, wind and waves.
- MoistSoilMgmtUnits (polygon) – Areas or units designed to enhance wetland habitat through the use of water containment, water supply and water control structures.
- Notch (point) – A notch in a riverine flow structure (typically a dike) used to control the flow of water into and out of the low flow area.
- PumpStation (point) – A location where a pump is located to pump water from the interior of a levee.
- Revetment (polygon) – An area faced with stone or other material along the shoreline designed to stabilize the bank or protect it from erosion.
- RiverineFlowStructureLine (line) – A natural or man-made flow (or sediment) control structure in a water course or water body such as a dike or weir. This feature should not be used to model a levee.
- RiverineFlowStructure (polygon) – A natural or man-made flow (or sediment) control structure in a water course or water body such as a dike or weir. This feature should not be used to model a levee.
- WaterControlStructure (point) – A structure designed to allow flow through an embankment.
- Well (point) – A shaft dug or drilled into the Earth for the purpose of extracting fluids from the subsurface, collecting environmental samples, injecting fluids into the subsurface or extracting contamination or other impurities from the subsurface.
- Wetland (polygon) – Geospatial feature composed of one unique wetland class (no complexes) used to determine USACE jurisdiction and regulatory impacts.

5 Dredging Event

5.1 Spatial & Geometric Definition

- Geometry Type: Polygon (Dredge_Event_Area).
- Spatial Delineation: Extents of authorized aquatic cut prisms within backwater lakes, secondary chutes, side channels, access cuts, and sediment deflection basins.
- Vertical Bounds: Existing hydrographic bathymetric profile down to engineered project design grade, including allowable design overdepth tolerance (+0.5 to +1.0 ft).
- Primary Restoration & Engineering Objectives Overwintering Fish Refugia: Deepen degraded shallow backwaters to restore minimum safe thermal depths (8.0 to 10.0 ft) beneath winter river ice cover.
- Riverine Velocity Dissipation: Provide low-velocity resting basins (<0.05 ft/s) protected from cold, turbulent main-stem drafts.
- Aquatic Habitat Longevity: Calculate design dredge depths using long-term historic bathymetric sedimentation rates to guarantee a minimum 25- to 50-year biological lifespan before re-shoaling.

Typical Cross-Sections, Dimensions & Sizing Criteria

- Bottom Widths: Primary fish cuts span 50 to 150 ft; deep winter holes span 100 to 300 ft.
- Side Slopes: Typically specified at 1V:3H to 1V:5H in unconsolidated alluvial sands, and 1V:6H to 1V:10H in soft cohesive lacustrine silts.
- Target Water Depths: Maintained at 8.0 to 12.0 ft below flat low navigation pool elevation.

Construction & Implementation Methodology

- Hydraulic Cutterhead Dredging: Slurry pumping (12 to 24 in pipelines) utilized when material is targeted for long-distance transport into confined disposal facilities (CDFs) or island containment berms.
- Mechanical Clamshell / Excavator Dredging: Barge-mounted long-reach excavators and scows utilized when sediments are coarse sand or placed immediately along adjacent island cores.
- Turbidity Controls: Silt curtains, hydraulic velocity dissipators, and continuous down-drift total suspended solids (TSS) and dissolved oxygen (DO) monitoring.

Quality Control, Inspection & Long-Term Maintenance

- Pre- and Post-Dredge Hydrographic Surveys: Multi-beam and dual-frequency single-beam acoustic sonar runs tied to USACE River Gages and NAVD88 vertical datum.
- Long-Term Monitoring: Bathymetric transects established at 200-ft intervals, surveyed at 1-, 3-, 5-, and 10-year intervals post-construction to quantify sediment accretion rates.

5.2 Spatial Feature Attribute

Definition

Backwater dredging typically consists of dredging channels with fingers (dredged channels that extend out away from the main dredge cut). The depth and size (length and width) of the dredge cut depends on several site specific factors.

Purpose

Sediment is dredged to create depth diversity in the backwaters can be used to enhance aquatic areas with islands or terrestrial areas with increase topographic diversity, which promotes the growth of mast trees.

Applicability

Many Upper Mississippi River System (UMRS) backwaters have been degraded by excessive amounts of sediment emanating from the basin, tributaries, and mainstem sources. This degradation is in the form of loss of depth, poor sediment quality, poor water quality, and sediment resuspension that blocks light required by aquatic plants. Streambank erosion throughout the basin is another important source of sediment that fills the backwaters. (UMRR Design Handbook, 2012)

Notes

Restoration features now have a Program_UID field to distinguish between NESP and UMRR features. Aliases without program identification will be created for any field that currently has UMRR in the name, so they can be used for NESP or UMRR data.

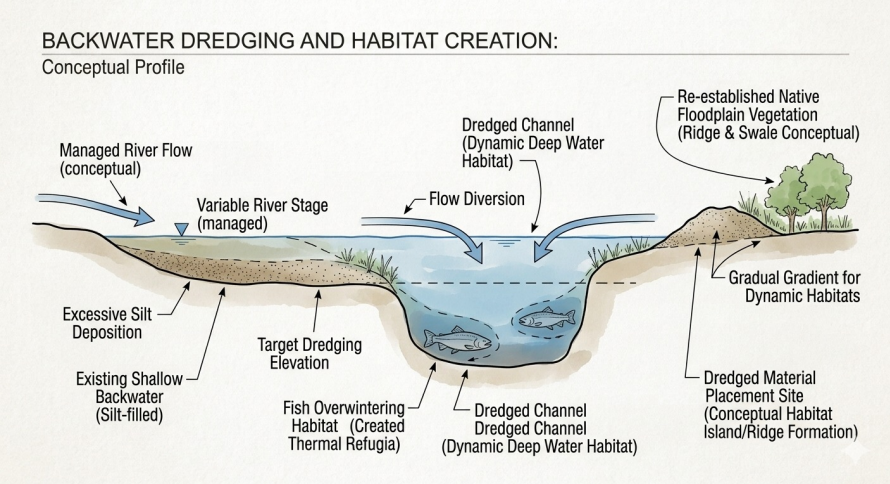
Attribute criteria for each field includes the following:

- Attribute Status – defines whether the field requires a value or if it is optional based on available information.
- Phases – indicates the earliest time in the project lifecycle values for this attribute as available.
- Digitizing requirements – indicates when the field value should be entered in the attribute table.

Geometry Note: Polygon nodes should have z values; elevations should be taken from as-built survey or design documents.

DredgingEvent

(polygon) – The extent of the area subject to dredging action. (area of material removal)



5.3

5.4 Feature Attribute Criteria

#

Attribute

Attribute Status

Phase

Digitizing Requirements

1

Program_UID

Required

Feasibility

At feature creation

2

PROJECT_ID

Required

Feasibility

At feature creation

3

Type

Required

Feasibility

At feature creation

4

Subtype

Required

Feasibility

At feature creation

5

RestorFeatStatus

Optional

Design

At feature creation

6

sdsFeatureName

Optional

Design

As required and available

7

sdsFeatureDescription

Optional

Design

As required and available

8

RestorFeatVdatum

Optional

Design

As required and available

9

DredgeMethod

Optional

Design

As required and available

10

VolumeCY

Optional

Construction

As required and available

11

CutWidthFT

Optional

Construction

As required and available

12

CutLengtFT

Optional

Construction

As required and available

13

CutDepthFT

Optional

Construction

As required and available

14

sdsMetaDataID

Optional

Construction

As required and available

15

RestorFeatIDPK

Optional

Construction

As required and available

16

sdsID

Optional

Construction

As required and available

17

mediaIDFK

Optional

Construction

As required and available

5.5 Parameters

Program_UID

Designate program NESP or UMRR.

Project_ID (alias: PROJECT_ID)

Foreign key to the unique NESP/UMRR project ID.

Type (alias)

A broad classification used to group NESP/UMRR restoration features into categories of similar function.

Value

Description

Dredging

Area subject to dredging action.

SubType (alias: Subtype)

A classification used to group NESP/UMRR restoration features into specific design feature categories.

Value

Description

Channel Dredging

Removal of sediment to create channels to redirect flow or provide equipment access to construction areas

Deep hole dredging

Removal of sediment, using a variety of methods, to create backwater lakes generally to increase fish habitat

Sediment basin dredging

Removal of sediment from or to create sediment basins

Geo-textile fill dredging

Removal of sediment to fill geo-textile tubes

Borrow

Removal of sediment from an area to be used for a specific purpose in another area.

RestorFeatStatus

A description of the status of the individual restoration. It also includes the party responsible for the individual feature.

Value

Description

USACE Planned

Feature is in a USACE planning stage.

USACE Built

Feature has been built by USACE. If a feature has been planned and built by USACE, once constructed, it will always be considered USACE built even if it has been repaired under later project stages.

USACE Enhanced

The function of an existing feature that was not built by USACE has been enhanced by a feature that was USACE built or repaired.

USACE Repaired

An existing feature that was not built by USACE has been directly improved or repaired.

Other Planned

A feature that is planned by another organization or government agency.

Other Built

Feature has been built by another organization or government agency and is an integral part of the UMRR project.

Other Enhanced

The function of an existing feature, that was built by another organization or government agency, has been enhanced by a feature that was built or repaired by a non-USACE organization or government agency.

Other Repaired

An existing feature that has been directly improved or repaired by another organization or government agency.

sdsFeatureName

The common name of the feature.

sdsFeatureDescription

A narrative describing the feature.

RestorFeatVdatum

Vertical datum of elevations recorded in attribute table or used as feature z values.

Value

Description

MSL 1912

Mean Sea Level of 1912

NGVD 1929

National Geodetic Vertical Datum of 1929

NAVD 1988

North American Vertical Datum of 1988

Other

Other, note datum in sdsFeatureDescription

Unknown

Unknown vertical datum

DredgeMethod

Method used to dredge.

Value

Description

Mechanical

Dredge method using mechanical equipment

Hydraulic

Dredge method using hydraulic equipment

High Solids

Dredge method used specifically for filling geotextile tubes

Unknown

Unknown

VolumeCY

Volume of material removed in cubic yards.

CutWidthFT

Width of dredge cut.

CutLengthFT

Length of dredge cut.

CutDepthFT

Average depth of dredge cut.

sdsMetaDataID

References attached metadata.

RestorFeatIDPK

References attached media.

sdsID

References attached media.

mediaIDFK

References attached media.

6 Embankment

6.1 Spatial & Geometric Definition

- Geometry Type: Polygon (Embankment_Footprint).
- Spatial Delineation: Terrestrial earthen footprint bounded by the outer intersection lines of outer fill slopes with native subgrade alluvium.
- Functional Roles: Isolating moist soil units, containing perimeter dredged placement cells, protecting wetlands from catastrophic river flood surges, and providing heavy maintenance vehicle access.

Primary Restoration & Engineering Objectives

- Hydraulic Isolation: Retain or exclude river floodwaters to facilitate independent moisture and pool-depth management.
- Controlled Overtopping Resistance: Engineered to withstand controlled, non-catastrophic submergence during 10- to 25-year flood recurrence events.
- Containment Integrity: Form stable retention dikes capable of impounding fine dredged hydraulic slurries without slope shear failures.

Typical Cross-Sections, Dimensions & Sizing Criteria

- Crest Width: 10.0 to 14.0 ft for maintenance roadways; 8.0 ft for non-vehicular perimeter berms.
- Side Slopes: Interior slopes at 1V:3H to 1V:4H; riverward exterior slopes at 1V:4H to 1V:6H to mitigate wind-wave runup.
- Structural Core: Central compacted low-permeability cohesive clay core (CL/CH) with a minimum 3.0-ft bottom cutoff trench keyed into native impermeable strata.
- Freeboard: Minimum 2.0 to 3.0 ft above maximum managed impoundment pool stage.

Construction & Implementation Methodology

- Subgrade Preparation: Stripping topsoil and organic muck (6 to 18 in); proof-rolling subgrade with heavy vibratory compactors.
- Lift Placement & Compaction: Clay and silty-clay fill spread in loose horizontal lifts of 8.0 in maximum, compacted to 95% of Standard Proctor Maximum Dry Density (ASTM D698) within ±2% of optimum moisture.
- Slope Armoring: Class I to Class II rock riprap placed over non-woven geotextile on exterior wave faces; native perennial root mats established on interior slopes.

Quality Control, Inspection & Long-Term Maintenance

- Geotechnical Monitoring: Piezometer installations along toe lines to monitor phreatic water lines and internal seepage gradients.
- Routine Inspection: Quarterly inspections for burrowing animals (muskrats/beavers), desiccation cracking, mechanical rutting, and woody tree encroachment along levee toes.

6.2 Spatial Feature Attribute

Definition

An embankment is a raised earthen structure constructed to control or contain water, provide flood protection, or create topographic diversity within floodplain ecosystems.

Purpose

Embankment systems are used to provide adequate flood protection, manage water levels, and shape floodplain topography. They can isolate management units, guide or restrict flows, protect infrastructure or habitats from inundation or erosion, and create elevated or sheltered areas that support diverse plant and wildlife communities.

Applicability

Embankments are applicable wherever control of water levels or patterns of inundation is needed in Upper Mississippi River System (UMRS) restoration projects. Typical uses include forming the boundaries of moist soil management units, creating berms around backwater lakes or wetlands, protecting restoration features from unwanted flooding or erosion, and shaping topographic relief to improve habitat diversity. (UMRR Design Handbook, 2012)

Notes

Restoration features now have a Program_UID field to distinguish between NESP and UMRR features. Aliases without program identification will be created for any field that currently has UMRR in the name, so they can be used for NESP or UMRR data.

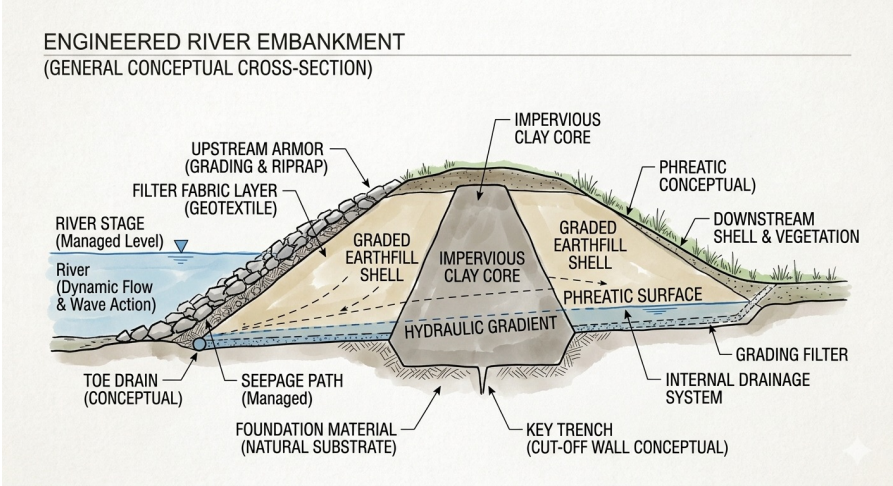
Attribute criteria for each field includes the following:

- Attribute Status – defines whether the field requires a value or if it is optional based on available information.
- Phases – indicates the earliest time in the project lifecycle values for this attribute are available.
- Digitizing requirements – indicates when the field value should be entered in the attribute table.

Geometry Note: Elevations should be taken from as-built survey or design documents.

Embankment

(polygon) – Embankments are raised earthen structures for the purpose of controlling water or creating topographic diversity in floodplain ecosystems.



6.3

6.4 Feature Attribute Criteria

#

Attributes

Attribute Status

Phase

Digitizing Requirement

1

Program_UID

Required

Feasibility

At feature creation

2

PROJECT_ID

Required

Feasibility

At feature creation

3

Type

Required

Feasibility

At feature creation

4

Subtype

Required

Feasibility

At feature creation

5

RestorFeatStatus

Required

Design

At feature creation

6

sdsFeatureName

Optional

Design

As required and available

7

sdsFeatureDescription

Optional

Design

As required and available

8

RestorFeatVdatum

Optional

Design

As required and available

9

CrestWidthFT

Optional

Construction

As required and available

10

EnbankmentMaterial

Optional

Construction

As required and available

11

EnbankmentLenghtFT

Optional

Construction

As required and available

12

EmbankmentSideSlope

Optional

Construction

As required and available

13

sdsMetaDataID

Optional

Construction

As required and available

14

RestorFeatIDPK

Optional

Construction

As required and available

15

sdsID

Optional

Construction

As required and available

16

mediaIDFK

Optional

Construction

As required and available

6.5 Parameters

Program_UID

Designate program NESP or UMRR.

Project_ID (alias: PROJECT_ID)

Foreign key to the unique NESP/UMRR project ID.

Type (alias)

A broad classification used to group NESP/UMRR restoration features into categories of similar function.

> Note: Specific coded values for Type were not provided in the source text for Embankment.
> In many implementations, Embankments fall under Floodplain Restoration or Water Level Management.

SubType (alias: Subtype)

A classification used to group NESP/UMRR restoration features into specific embankment design feature categories.

> Note: SubType coded values were not provided in the source text.
> Example values might include Exterior Berm, Interior Berm, Levee Segment, or Training Embankment, depending on project conventions.

RestorFeatStatus

A description of the status of the individual restoration. It also includes the party responsible for the individual feature.

Value

Description

USACE Planned

Feature is in a USACE planning stage.

USACE Built

Feature has been built by USACE. If a feature has been planned and built by USACE, once constructed, it will always be considered USACE built even if it has been repaired under later project stages.

USACE Enhanced

The function of an existing feature that was not built by USACE has been enhanced by a feature that was USACE built or repaired.

USACE Repaired

An existing feature that was not built by USACE has been directly improved or repaired.

Other Planned

A feature that is planned by another organization or government agency.

Other Built

Feature has been built by another organization or government agency and is an integral part of the UMRR project.

Other Enhanced

The function of an existing feature, that was built by another organization or government agency, has been enhanced by a feature that was built or repaired by a non-USACE organization or government agency.

Other Repaired

An existing feature that has been directly improved or repaired by another organization or government agency.

sdsFeatureName

The common name of the feature.

sdsFeatureDescription

A narrative describing the feature.

RestorFeatVdatum

Vertical datum of elevations recorded in the attribute table or used as feature z values.

Value

Description

MSL 1912

Mean Sea Level of 1912.

NGVD 1929

National Geodetic Vertical Datum of 1929.

NAVD 1988

North American Vertical Datum of 1988.

Other

Other; note datum in sdsFeatureDescription.

Unknown

Unknown vertical datum.

CrestWidthFT

Width of the embankment top or crest in feet.

EnbankmentMaterial

Primary material used to construct the embankment.

Value

Description

Sand

Sand material.

Clay

Clay material.

Sand with clay core

Embankment composed primarily of sand with a central clay core.

Sand with clay cap

Embankment composed primarily of sand with a clay cap.

Earthen

Mixture of sand and clay, used when material type is not specified in design or as-built drawings.

EnbankmentLenghtFT

Length of the embankment in feet (note: field name includes a typographical error “Lenght” retained for schema consistency).

EmbankmentSideSlope

Side slope of the embankment, typically expressed as a horizontal-to-vertical ratio (e.g., 3H:1V). May be stored as text or numeric representation per data model convention.

sdsMetaDataID

References attached metadata.

RestorFeatIDPK

Primary key used to uniquely identify the restoration feature record.

sdsID

Identifier used to link the feature to supporting SDS (spatial data system) records or related datasets.

mediaIDFK

Foreign key used to reference associated media (e.g., design drawings, cross sections, as-built surveys, photographs) documenting the embankment.

7 Embankment Centerline

7.1 Spatial & Geometric Definition

- Geometry Type: Polyline (Embankment_Centerline_Line).
- Spatial Delineation: Longitudinal survey axis running along the geometric centroid of the top crest of the embankment prism.
- Data Fields: Linear stationing (0+00), top-of-crest design profile elevation, horizontal curve radii, design freeboard height, and crest width domain codes.

Primary Restoration & Engineering Objectives

- Engineering Geometry Control: Serves as the master horizontal and vertical control baseline for all construction layout, cross-sectional cut/fill grading, and post-settlement monitoring.
- Profile Consistency: Establishes the uniform, non-dip crest grade to ensure overtopping occurs exclusively at designated, armored spillway notches rather than across unprotected earthen crest segments.

Typical Cross-Sections, Dimensions & Sizing Criteria

- Survey Baseline: Stationing established at 100-ft linear increments with permanent control monuments positioned at Points of Curvature (PC) and Points of Tangency (PT).
- Grade Tolerances: Construction grade tolerance along centerline is strict: -0.00 ft/$+0.20$ ft from contract design profile.

Construction & Implementation Methodology

- RTK-GPS & Total Station Layout: Field staked with high-precision RTK-GPS tied directly into national State Plane Coordinates (NAD83) and NAVD88 elevation benchmarks.
- Machine Control Integration: Centerline coordinate and profile strings loaded into 3D GPS machine-guidance earthmoving excavators and motor graders.

Quality Control, Inspection & Long-Term Maintenance

- Settlement Monitoring: Permanent brass survey disks embedded in concrete surface monuments every 500 ft along the centerline, surveyed annually to track long-term primary consolidation of underlying river silts.

7.2 Spatial Feature Attribute

Definition

A linear embankment feature represents the centerline of a raised earthen structure constructed to control or contain water, provide flood protection, or create topographic diversity within floodplain ecosystems.

Purpose

Embankment systems are used to provide adequate flood protection, manage water levels, and shape floodplain topography. Linear representation of the embankment centerline supports design, hydraulic analysis, and asset management by defining the primary alignment of the structure. Embankments can isolate management units, guide or restrict flows, protect infrastructure or habitats from inundation or erosion, and create elevated or sheltered areas that support diverse plant and wildlife communities.

Applicability

Linear embankment features are applicable wherever centerline representation of raised earthen structures is needed for planning, design, modeling, or operations within Upper Mississippi River System (UMRS) restoration projects. This includes levees, berms around MSMUs and backwater lakes, training embankments, and other floodplain earthworks that control water or create topographic variation. (UMRR Design Handbook, 2012)

Notes

Restoration features now have a Program_UID field to distinguish between NESP and UMRR features. Aliases without program identification will be created for any field that currently has UMRR in the name, so they can be used for NESP or UMRR data.

Attribute criteria for each field includes the following:

- Attribute Status – defines whether the field requires a value or if it is optional based on available information.
- Phases – indicates the earliest time in the project lifecycle values for this attribute are available.
- Digitizing requirements – indicates when the field value should be entered in the attribute table.

Geometry Note: Line should be 3D; elevations should be taken from as-built survey or design documents.

Embankment

(line) – The centerline of a linear, raised earthen structure, for the purpose of controlling water or creating topographic diversity in floodplain ecosystems.

7.3 Feature Attribute Criteria

#

Attributes

Attribute Status

Phase

Digitizing Requirement

1

Program_UID

Required

Feasibility

At feature creation

2

PROJECT_ID

Required

Feasibility

At feature creation

3

Type

Required

Feasibility

At feature creation

4

Subtype

Required

Feasibility

At feature creation

5

RestorFeatStatus

Required

Design

At feature creation

6

sdsFeatureName

Optional

Design

As required and available

7

sdsFeatureDescription

Optional

Design

As required and available

8

RestorFeatVdatum

Optional

Design

As required and available

9

CrestWidthFT

Optional

Construction

As required and available

10

EnbankmentMaterial

Optional

Construction

As required and available

11

EnbankmentLenghtFT

Optional

Construction

As required and available

12

EmbankmentSideSlope

Optional

Construction

As required and available

13

sdsMetaDataID

Optional

Construction

As required and available

14

RestorFeatIDPK

Optional

Construction

As required and available

15

sdsID

Optional

Construction

As required and available

16

mediaIDFK

Optional

Construction

As required and available

7.4 Parameters

Program_UID

Designate program NESP or UMRR.

Project_ID (alias: PROJECT_ID)

Foreign key to the unique NESP/UMRR project ID.

Type (alias)

A broad classification used to group NESP/UMRR restoration features into categories of similar function.

> Note: Specific coded values for Type were not provided for the line Embankment. Embankments are commonly associated with Floodplain Restoration or Water Level Management in project schemas.

SubType (alias: Subtype)

A classification used to group NESP/UMRR restoration features into specific embankment design feature categories.

> Note: SubType coded values were not provided for the line Embankment.
> Example values might include Exterior Berm, Interior Berm, Levee Centerline, or Training Embankment, aligned with the polygon Embankment subtypes.

RestorFeatStatus

A description of the status of the individual restoration. It also includes the party responsible for the individual feature.

Value

Description

USACE Planned

Feature is in a USACE planning stage.

USACE Built

Feature has been built by USACE. If a feature has been planned and built by USACE, once constructed, it will always be considered USACE built even if it has been repaired under later project stages.

USACE Enhanced

The function of an existing feature that was not built by USACE has been enhanced by a feature that was USACE built or repaired.

USACE Repaired

An existing feature that was not built by USACE has been directly improved or repaired.

Other Planned

A feature that is planned by another organization or government agency.

Other Built

Feature has been built by another organization or government agency and is a integral part of the UMRR project.

Other Enhanced

The function of an existing feature, that was built by another organization or government agency, has been enhanced by a feature that was built or repaired by a non-USACE organization or government agency.

Other Repaired

An existing feature that has been directly improved or repaired by another organization or government agency.

sdsFeatureName

The common name of the feature.

sdsFeatureDescription

A narrative describing the feature, including alignment, purpose, and notable design characteristics.

RestorFeatVdatum

Vertical datum of elevations recorded in the attribute table or used as feature z values for the 3D line.

Value

Description

MSL 1912

Mean Sea Level of 1912.

NGVD 1929

National Geodetic Vertical Datum of 1929.

NAVD 1988

North American Vertical Datum of 1988.

Other

Other; note datum in sdsFeatureDescription.

Unknown

Unknown vertical datum.

EnbankmentMaterial

Primary material used to construct the embankment, identified for the centerline segment.

Value

Description

Sand

Sand material.

Clay

Clay material.

Sand with clay core

Embankment composed primarily of sand with a central clay core.

Sand with clay cap

Embankment composed primarily of sand with a clay cap.

Earthen

Some mixture of sand and clay, used when material type is not specified in design or as-built drawings.

CrestWidthFT

Width of the embankment top or crest in feet (associated with the line segment).

EnbankmentLenghtFT

Length of the embankment in feet for the represented line feature (note: field name includes typographical error “Lenght” retained for schema consistency).

EmbankmentSideSlope

Side slope of the embankment, expressed as a horizontal-to-vertical ratio (e.g., 3H:1V), applicable to the represented centerline segment.

sdsMetaDataID

References attached metadata.

RestorFeatIDPK

Primary key used to uniquely identify the restoration feature record.

sdsID

Identifier used to link the feature to supporting SDS (spatial data system) records or related datasets.

mediaIDFK

Foreign key used to reference associated media (e.g., design profiles, cross-sections, levee certification documents, as-built surveys, photographs) documenting the embankment.

8 Fish Structure

8.1 Spatial & Geometric Definition

- Geometry Type: Point or Polygon (Fish_Structure_Location).
- Spatial Delineation: Precise spatial coordinates or bounding footprints of submerged structural habitat units embedded in backwater cuts, secondary channels, and plunge basins.
- Component Typology: Anchored large woody debris (LWD) complexes, rootwad revetments, brush bundles, boulder clusters, and wooden crib reefs.

Primary Restoration & Engineering Objectives

- Cover & Velocity Shelters: Create localized zones of zero velocity (0.0 ft/s) alongside predatory feeding seams for gamefish species (largemouth bass, bluegill, black crappie, walleye).
- Macroinvertebrate Substrate: Provide durable, non-silt-covered hard biological surfaces for net-spinning caddisflies, mayflies, and epilithic periphyton.
- Thermal Microhabitats: Maintain localized deep cover within overwintering basins, preventing schooling fish from expending critical winter energy reserves.

Typical Cross-Sections, Dimensions & Sizing Criteria

- Boulder Clusters: Groups of 3 to 7 interlocking Class IV–V boulders (d 50 =24 to 36 in), placed in staggered diamond formations spanning 15×25 ft footprints.
- Rootwad / Tree Deflectors: Hardwood tree trunks (18 in diameter) with intact root fans (6 to 10 ft diameter), projecting 10 to 20 ft from channel margins.
- Ballast Requirements: Stone ballast sizing calculated to offset submerged wood buoyancy by a factor of safety FS 1.75 (1.5 to 2.0 tons of rock per full-size tree bole).

Construction & Implementation Methodology

- Barge-Mounted Crane Placement: Lowered into aquatic cuts using cable chokers or mechanical clamshells.
- Anchoring Systems: Hardwoods secured to riverbeds using duckbill earth anchors, helical steel ground screws, or heavy galvanized aircraft cables (3/8 in diameter) clamped through drilled stone anchors.

Quality Control, Inspection & Long-Term Maintenance

- Underwater Acoustic Surveys: High-resolution multi-beam side-scan sonar runs conducted at 2- and 5-year intervals to verify structural presence, orientation, and absence of silt smothering.
- Biological Electrofishing: Annual fisheries index surveys comparing fish density at structure sites against un-structured control reaches.

8.2 Spatial Feature Attributes

Definition

FishStructure features are engineered structures designed to improve fish habitat or allow improved fish movement through the river system by modifying hydraulics, providing cover, or circumventing in-channel barriers.

Purpose

Improving fish habitat and facilitating fish movement through the river system are critical steps in maintaining the health, biodiversity, and sustainability of the UMR ecosystem. These efforts support a wide range of species, restore ecological functions, enhance fish populations, and contribute to the local economy through sustainable fisheries and recreation.

Applicability

Fish structures are applicable wherever in-channel barriers, degraded habitat, or limited connectivity constrain fish movement or habitat quality within Upper Mississippi River System (UMRS) restoration projects. They are typically used at dams, diversion structures, constricted channels, and along eroding banks or simplified shorelines to restore passage, create refuge, and diversify hydraulic and substrate conditions. (UMRR Design Handbook, 2012)

Notes

Restoration features now have a Program_UID field to distinguish between NESP and UMRR features. Aliases without program identification will be created for any field that currently has UMRR in the name, so they can be used for NESP or UMRR data.

Attribute criteria for each field includes the following:

- Attribute Status – defines whether the field requires a value or if it is optional based on available information.
- Phases – indicates the earliest time in the project lifecycle values for this attribute are available.
- Digitizing requirements – indicates when the field value should be entered in the attribute table.

Geometry Note: Points should have well top elevation as Z value if available.

FishStructure

(polygon) – An engineered structure designed to improve fish habitat or allow improved fish movement through the river system.

8.3 Feature Attribute Criteria

#

Attributes

Attribute Status

Phase

Digitizing Requirement

1

Program_UID

Required

Feasibility

At feature creation

2

PROJECT_ID

Required

Feasibility

At feature creation

3

Type

Required

Feasibility

At feature creation

4

Subtype

Required

Feasibility

At feature creation

5

RestorFeatStatus

Required

Design

At feature creation

6

sdsFeatureName

Optional

Design

As required and available

7

sdsFeatureDescription

Optional

Design

As required and available

8

RestorFeatVdatum

Optional

Design

As required and available

9

TargetFishSpecies

Optional

Design

As required and available

10

StartStation

Optional

Construction

As required and available

11

EndStation

Optional

Construction

As required and available

12

sdsMetaDataID

Optional

Construction

As required and available

13

RestorFeatIDPK

Optional

Construction

As required and available

14

sdsID

Optional

Construction

As required and available

15

mediaIDFK

Optional

Construction

As required and available

8.4 Parameters

Program_UID

Designate program NESP or UMRR.

Project_ID (alias: PROJECT_ID)

Foreign key to the unique NESP/UMRR project ID.

Type (alias)

A broad classification used to group NESP/UMRR restoration features into categories of similar function.

Value

Description

Channel Modifications

Measures taken to train the main channel of a river and to modify secondary channels.

SubType (alias: Subtype)

A classification used to group NESP/UMRR restoration features into specific fish structure design feature categories.

Value

Description

Fish LUNKERS

Engineered overhanging-bank structure that provides fish habitat and bank stabilization, generally about 8 ft long, 1–2 ft tall and 3 ft deep, constructed of hardwood, concrete, or plastic lumber.

Fish bypass channel

Channel created to circumvent in-channel barriers to natural fish movement.

Fish ladder

Engineered step or pool-and-weir structure created to circumvent in-channel barriers to natural fish movement.

Natural boulder ramp

Short, steep ramp comprised of boulders created to circumvent in-channel barriers to natural fish movement.

RestorFeatStatus

A description of the status of the individual restoration. It also includes the party responsible for the individual feature.

Value

Description

USACE Planned

Feature is in a USACE planning stage.

USACE Built

Feature has been built by USACE. If a feature has been planned and built by USACE, once constructed, it will always be considered USACE built even if it has been repaired under later project stages.

USACE Enhanced

The function of an existing feature that was not built by USACE has been enhanced by a feature that was USACE built or repaired.

USACE Repaired

An existing feature that was not built by USACE has been directly improved or repaired.

Other Planned

A feature that is planned by another organization or government agency.

Other Built

Feature has been built by another organization or government agency and is an integral part of the UMRR project.

Other Enhanced

The function of an existing feature, that was built by another organization or government agency, has been enhanced by a feature that was built or repaired by a non-USACE organization or government agency.

Other Repaired

An existing feature that has been directly improved or repaired by another organization or government agency.

sdsFeatureName

The common name of the feature.

sdsFeatureDescription

A narrative describing the feature, including design intent, structural details, and operational assumptions.

RestorFeatVdatum

Vertical datum of elevations recorded in the attribute table or used as feature z values.

Value

Description

MSL 1912

Mean Sea Level of 1912.

NGVD 1929

National Geodetic Vertical Datum of 1929.

NAVD 1988

North American Vertical Datum of 1988.

Other

Other; note datum in sdsFeatureDescription.

Unknown

Unknown vertical datum.

TargetFishSpecies

Specific fish species or species groups the structure is intended to benefit (e.g., pallid sturgeon, paddlefish, centrarchids). May include one or more species names or functional groups.

StartStation

Beginning station of the feature, typically referenced to a project or river engineering baseline (e.g., 10+00).

EndStation

Ending station of the feature along the same baseline.

sdsMetaDataID

References attached metadata (e.g., design reports, hydraulic modeling files, biological objectives).

RestorFeatIDPK

Primary key used to uniquely identify the restoration feature record.

sdsID

Identifier used to link the feature to supporting SDS (spatial data system) records or related datasets.

mediaIDFK

Foreign key used to reference associated media (e.g., design drawings, plans and profiles, construction photographs, as-built surveys) documenting the fish structure.

9 Flora Planting Seeding

9.1 Spatial & Geometric Definition

- Geometry Type: Polygon (Flora_Planting_Area).
- Spatial Delineation: Managed zones receiving revegetation treatments across restored island crests, moist soil units, dredge disposal sites, and riparian forest floodplains.
- Vegetation Types: Hard-mast bottomland hardwoods, native emergent wetland tubers/plugs, pollinator forb mixes, and deep-rooted native warm-season prairie grasses.

Primary Restoration & Engineering Objectives

- Soil Bio-Stabilization: Rapid root-mesh development to anchor sandy and fine-grained dredged placement materials against wave-wash and flood shear stress.
- Wildlife Mast Production: Re-establish critical hard-mast forest species (swamp white oak, pin oak, shellbark hickory, pecan) displaced by altered post-dam hydrologic regimes.
- Invasive Species Suppression: Establish continuous native vegetative cover to prevent colonization by monotypic reed canary grass (Phalaris arundinacea).

Typical Specifications, Densities & Seed Mixes

- Hard-Mast Tree Stock: Root Production Method (RPM) or bare-root containerized saplings (1.0-gal to 3.0-gal pots, 0.5 in caliper). Planted at densities of 200 to 300 stems/acre on 12×12 ft to 15×15 ft grids.
- Riparian Seed Mixes: Native cool- and warm-season grasses (Virginia wildrye, river bulrush, cordgrass, bluejoint grass) broadcast at 10 to 20 lbs Pure Live Seed (PLS)/acre.
- Emergent Wetlands: Bare-root tubers/plugs of Sagittaria latifolia and Scirpus fluviatilis installed along littoral margins (0.5 to 2.0 ft water depth) on 3-ft centers.

Construction & Implementation Methodology

- Site Preparation: Multi-pass forestry mowing followed by aquatic-approved glyphosate herbicide treatments to eradicate invasive root mats.
- Shelter Tube Installation: Hardwood saplings protected with 48- to 60-in rigid, ventilated photodegradable polyethylene tree shelters staked with rot-resistant white oak stakes to eliminate deer browse and rodent girdling.

Quality Control, Inspection & Long-Term Maintenance

- Survival Censuses: Fixed-radius inventory plots (0.1 acre) evaluated at months 12, 24, and 36 post-planting. Target standard is 80% survival density.
- Herbicide Touch-Ups: Backpack spot-spraying or mechanical brush-cutting around tree collars for 3 consecutive growing seasons until sapling crowns overtop competing weeds.

9.2 Spatial Feature Attribute

Definition

Flora Planting or Seeding is a floodplain restoration measure in which vegetation is intentionally planted or seeded to improve natural floodplain function and habitat.

Purpose

Flora planting or seeding is used to establish or enhance native plant communities, stabilize soils, increase structural and species diversity, and provide food and cover for wildlife. These actions support long-term habitat goals, improve resilience to flooding and disturbance, and help restore natural floodplain processes such as nutrient cycling and sediment retention.

Applicability

Flora planting or seeding is applicable wherever vegetation needs to be established, enhanced, or converted within Upper Mississippi River System (UMRS) restoration projects. Typical applications include revegetating disturbed areas, restoring floodplain forests, wetlands, and prairies, improving shoreline and island stability, and enhancing terrestrial and aquatic edge habitats. (UMRR Design Handbook, 2012)

Notes

Restoration features now have a Program_UID field to distinguish between NESP and UMRR features. Aliases without program identification will be created for any field that currently has UMRR in the name, so they can be used for NESP or UMRR data.

Attribute criteria for each field includes the following:

- Attribute Status – defines whether the field requires a value or if it is optional based on available information.
- Phases – indicates the earliest time in the project lifecycle values for this attribute are available.
- Digitizing requirements – indicates when the field value should be entered in the attribute table.

Geometry Note: No Z values are needed for this feature.

FloraPlantingOrSeeding

(polygon) – A location where vegetation is planted or seeded.

9.3 Feature Attribute Criteria

#

Attributes

Attribute Status

Phase

Digitizing Requirement

1

Program_UID

Required

Feasibility

At feature creation

2

PROJECT_ID

Required

Feasibility

At feature creation

3

Type

Required

Feasibility

At feature creation

4

Subtype

Required

Feasibility

At feature creation

5

RestorFeatStatus

Required

Design

At feature creation

6

sdsFeatureName

Optional

Design

As required and available

7

sdsFeatureDescription

Optional

Design

As required and available

8

RestorFeatVdatum

Optional

Design

As required and available

9

PlantingDate

Optional

Design

As required and available

10

DominantSpecies1

Optional

Construction

As required and available

11

DominantSpecies2

Optional

Construction

As required and available

12

DominantSpecies3

Optional

Construction

As required and available

13

MastPlanting

Optional

Construction

As required and available

14

sdsMetaDataID

Optional

Construction

As required and available

15

RestorFeatIDPK

Optional

Construction

As required and available

16

sdsID

Optional

Construction

As required and available

17

mediaIDFK

Optional

Construction

As required and available

9.4 Parameters

Program_UID

Designate program NESP or UMRR.

Project_ID (alias: PROJECT_ID)

Foreign key to the unique NESP/UMRR project ID.

Type (alias)

A broad classification used to group NESP/UMRR restoration features into categories of similar function.

Value

Description

Shoreline Protection

Measures taken to reduce erosion from banks and islands due to wind, wave, and ice action.

Floodplain Restoration

Measures taken to improve natural floodplain function and habitat.

SubType (alias: Subtype)

A classification used to group NESP/UMRR restoration features into specific planting or seeding design feature categories.

Value

Description

Tree planting

Planting of tree species in floodplain areas that have appropriate soil types and water regimes.

Wetland planting

Planting of wetland species in areas with suitable hydrology and soils to establish or enhance wetland habitat.

Prairie Planting

Planting of prairie species in floodplain areas with appropriate soil types and water regimes.

Floodplain Forest Planting

Planting or augmentation of native floodplain tree and shrub species to establish or restore forested floodplain habitat.

Timber Stand Improvement

Silvicultural treatments within existing forest stands (e.g., thinning, release, underplanting) to improve species composition, structure, and habitat quality.

Aquatic Vegetation

Planting or seeding of emergent, submergent, or floating-leaved aquatic vegetation to improve aquatic habitat and stabilize substrates.

RestorFeatStatus

A description of the status of the individual restoration. It also includes the party responsible for the individual feature.

Value

Description

USACE Planned

Feature is in a USACE planning stage.

USACE Built

Feature has been implemented (planted/seeded) by USACE. Once implemented by USACE, it remains classified as USACE Built even if later maintained or repaired.

USACE Enhanced

The function of an existing planting or habitat area not originally established by USACE has been enhanced by a USACE action.

USACE Repaired

An existing planting area not established by USACE has been directly improved or repaired by USACE (e.g., replanting after failure).

Other Planned

Planting/seeding feature planned by another organization or government agency.

Other Built

Planting/seeding feature implemented by another organization or government agency and integral to the UMRR project.

Other Enhanced

Function of an existing planting, built by another organization, enhanced by a non-USACE organization or government agency.

Other Repaired

Existing planting directly improved or repaired by another organization or government agency.

sdsFeatureName

The common or management name of the planting area.

sdsFeatureDescription

A narrative describing the planting or seeding feature, including objectives, methods, and any notable site conditions.

RestorFeatVdatum

Vertical datum of any elevations recorded in the attribute table (if used for related features).

Value

Description

MSL 1912

Mean Sea Level of 1912.

NGVD 1929

National Geodetic Vertical Datum of 1929.

NAVD 1988

North American Vertical Datum of 1988.

Other

Other; note datum in sdsFeatureDescription.

Unknown

Unknown vertical datum.

PlantingDate

The date the planting or seeding was carried out (or first completed).

DominantSpecies1

Dominant or co-dominant species planted or seeded within the feature. Percentage of the planting mix can be included as part of the value or in sdsFeatureDescription.

DominantSpecies2

Second dominant or co-dominant species planted or seeded. Percentage of the planting mix can be included.

DominantSpecies3

Third dominant or co-dominant species planted or seeded. Percentage of the planting mix can be included.

MastPlanting

Indicates whether the planting provides mast (nuts, acorns, berries, etc.) important to wildlife species.

Value

Description

Yes

Planting intended to provide mast for wildlife.

No

Planting not specifically intended as a mast-producing planting.

Unknown

Mast function not specified.

sdsMetaDataID

References attached metadata (e.g., seed mix specifications, planting plans, monitoring protocols).

RestorFeatIDPK

Primary key used to uniquely identify the restoration feature record.

sdsID

Identifier used to link the feature to supporting SDS (spatial data system) records or related datasets.

mediaIDFK

Foreign key used to reference associated media (e.g., photographs, planting plans, seed tags, monitoring reports) documenting the planting or seeding.

10 Flow Channel

10.1 Spatial & Geometric Definition

- Geometry Type: Polygon (Flow_Channel_Footprint).
- Spatial Delineation: Channel boundaries of excavated aquatic chutes, secondary conveyance corridors, or backwater flow-supply corridors reconnecting the main river to isolated interior wetlands.
- Hydraulic Classification: Continuous low-velocity flow channels, seasonal skimming channels, or flood pulse distributors.

Primary Restoration & Engineering Objectives

- Oxygenated Water Delivery: Introduce low volumes of oxygen-rich river water into stagnant interior backwaters to prevent winter and summer anoxic fish kills.
- Biological Corridor Connectivity: Provide passage routes for migratory riverine fish traveling between river feeding reaches and sheltered nursery lakes.
- Controlled Energy Conveyance: Route discharge without inducing bed scour, toe destabilization, or excessive sediment imports into fragile interior basins.

Typical Cross-Sections, Dimensions & Sizing Criteria

- Bottom Widths: 20 to 60 ft; side slopes stabilized at 1V:3H or flatter.
- Longitudinal Bed Slope: Very flat gradient (S 0.0002 to 0.0005) to constrain cross-sectional average velocities beneath critical erosive limits (v 1.0 to 1.5 ft/s).
- Inlet Control Sills: Hard-armored rock sills placed at the channel inlet set 0.5 to 1.5 ft above normal pool to block bedload sediment entry during low-water regimes.

Construction & Implementation Methodology

- Excavation Fleet: Low-ground-pressure (LGP) amphibious excavators or spud-barge mechanical dredge rigs working from downstream upstream to maintain positive gravity dewatering.
- Bank Armor Integration: Riprap placement along channel entrance mouths and sharp bendway curves subject to centrifugal shear forces.

Quality Control, Inspection & Long-Term Maintenance

- Acoustic Flow Gauging: Seasonal ADCP velocity transects to confirm flow splits match calibrated hydrodynamic model predictions (typically 1% to 5% of main river discharge).
- Debris Clearance: Annual post-flood inspection to clear lodged drift logs and alluvial sand wedges from channel entrances.

10.2 Spatial Feature Attribute

Definition

A FlowChannel is an excavated channel, created by mechanical or hydraulic means, intended to allow the flow of water across, into, or out of the project area or any subcomponent of a project area.

Purpose

Flow channels are constructed to direct, convey, or redistribute water in support of restoration objectives. They can be used to reconnect floodplains or backwaters, improve circulation, manage water levels, enhance habitat connectivity, and reduce stagnant or low-quality water conditions by providing designed flow paths.

Applicability

Flow channels are applicable wherever controlled or enhanced conveyance of water is needed within Upper Mississippi River System (UMRS) restoration projects. They are used in both terrestrial and aquatic settings to route water between project features (e.g., MSMUs, backwater lakes, side channels), restore or mimic natural flow patterns, and support habitat, sediment, and water quality objectives. (UMRR Design Handbook, 2012)

Notes

Restoration features now have a Program_UID field to distinguish between NESP and UMRR features. Aliases without program identification will be created for any field that currently has UMRR in the name, so they can be used for NESP or UMRR data.

Attribute criteria for each field includes the following:

- Attribute Status – defines whether the field requires a value or if it is optional based on available information.
- Phases – indicates the earliest time in the project lifecycle values for this attribute are available.
- Digitizing requirements – indicates when the field value should be entered in the attribute table.

Geometry Note: Polygon nodes should have z values; elevations should be taken from as-built survey or design documents.

FlowChannel

(polygon) – An excavated channel (via mechanical or hydraulic means) intended to allow for the flow of water across, into, or out of the project area or any subcomponent of a project area.

10.3 Feature Attribute Criteria

#

Attribute

Attribute Status

Phase

Digitizing Requirements

1

Program_UID

Required

Feasibility

At feature creation

2

PROJECT_ID (Alias)

Required

Feasibility

At feature creation

3

Type (Alias)

Required

Feasibility

At feature creation

4

Subtype (Alias)

Required

Feasibility

At feature creation

5

RestorFeatStatus

Required

Feasibility

At feature creation

6

sdsFeatureName

Optional

Design

As required and available

7

sdsFeatureDescription

Optional

Design

As required and available

8

RestorFeatVdatum

Optional

Design

As required and available

9

ConstructionMethod

Optional

Design

As required and available

10

ConstructionMethod

Optional

Design

As required and available

11

VolumeCY

Optional

Design

As required and available

12

CutWidthFT

Optional

Design

As required and available

13

CutLengthFT

Optional

Design

As required and available

14

CutDepthFT

Optional

Construction

As required and available

15

sdsMetaDataID

Optional

Construction

As required and available

16

RestorFeatIDPK

Optional

Construction

As required and available

17

sdsID

Optional

Construction

As required and available

18

mediaIDFK

Optional

Construction

As required and available

10.4 Parameters

Program_UID

Designate program NESP or UMRR.

Project_ID (alias: PROJECT_ID)

Foreign key to the unique NESP/UMRR project ID.

Type (alias: Type)

A broad classification used to group NESP/UMRR restoration features into categories of similar function.

Value

Description

Flow Channel

An excavated channel (via mechanical or hydraulic means) intended to allow for the flow of water across, into, or out of the project area or any subcomponent of a project area.

SubType (alias: Subtype)

A classification used to group NESP/UMRR restoration features into specific design feature categories.

Value

Description

Terrestrial

Removal of sediment to create channels to allow for flow of water across or through predominantly terrestrial areas.

Aquatic

Removal of sediment to create channels to allow for flow of water within predominantly aquatic areas.

RestorFeatStatus

A description of the status of the individual restoration. It also includes the party responsible for the individual feature.

Value

Description

USACE Planned

Feature is in a USACE planning stage.

USACE Built

Feature has been built by USACE. If a feature has been planned and built by USACE, once constructed, it will always be considered USACE built even if it has been repaired under later project stages.

USACE Enhanced

The function of an existing feature that was not built by USACE has been enhanced by a feature that was USACE built or repaired.

USACE Repaired

An existing feature that was not built by USACE has been directly improved or repaired.

Other Planned

A feature that is planned by another organization or government agency.

Other Built

Feature has been built by another organization or government agency and is an integral part of the UMRR project.

Other Enhanced

The function of an existing feature, that was built by another organization or government agency, has been enhanced by a feature that was built or repaired by a non-USACE organization or government agency.

Other Repaired

An existing feature that has been directly improved or repaired by another organization or government agency.

sdsFeatureName

The common name of the feature.

sdsFeatureDescription

A narrative describing the feature.

RestorFeatVdatum

Vertical datum of elevations recorded in the attribute table or used as feature z values.

Value

Description

MSL 1912

Mean Sea Level of 1912.

NGVD 1929

National Geodetic Vertical Datum of 1929.

NAVD 1988

North American Vertical Datum of 1988.

Other

Other; note datum in sdsFeatureDescription.

Unknown

Unknown vertical datum.

ConstructionMethod

Method used to remove sediment and construct the flow channel.

Value

Description

Mechanical

Removal method using mechanical equipment (e.g., excavators, draglines).

Hydraulic

Removal method using hydraulic equipment (e.g., cutterhead or plain suction dredges).

VolumeCY

Volume of material removed in cubic yards.

CutWidthFT

Width of the excavated channel in feet.

CutLengthFT

Length of the excavated channel in feet.

CutDepthFT

Average depth of the excavated channel in feet.

sdsMetaDataID

References attached metadata.

RestorFeatIDPK

Primary key used to uniquely identify the restoration feature record.

sdsID

Identifier used to link the feature to supporting SDS (spatial data system) records or related datasets.

mediaIDFK

Foreign key used to reference associated media (e.g., design drawings, dredging records, as-built plans, photographs) documenting the flow channel.

11 Island Centerline

11.1 Spatial & Geometric Definition

- Geometry Type: Polyline (Island_Crest_Line or Island_Baseline).
- Spatial Delineation: Centerline axis, design baseline, or high-crest ridge running continuously along the longitudinal alignment of a constructed island.
- Data Fields: Island segment ID, design crest profile grade, baseline stationing, revetment armoring class, and design overtopping threshold.

Primary Restoration & Engineering Objectives

- Survey Baseline Control: Serves as the primary reference vector from which cross-sectional design offset templates (perpendicular cross-sections) are generated and staked.
- Hydraulic Ridge Integrity: Tracks the continuous elevation of the island's primary windbreak and flood-deflection backbone.

Typical Specifications & Geometric Properties

- Stationing System: Formatted identically to civil alignments (0+00 to L), with stationing typically progressing from the downstream trailing tip to the upstream prow.
- Elevation Control: Tied strictly to the project vertical control network (NAVD88) with construction staking tolerances of ± 0.1 ft.

Construction & Implementation Methodology

- GPS Navigation Strings: Exported as civil design alignment files into real-time barge GPS tracking suites to guide placement derrick operators and bulldozer operators.

Quality Control, Inspection & Long-Term Maintenance

- Crest Profile Resurveys: Periodic RTK-GPS profile walks along the island line following major flood inundation cycles to calculate post-consolidation subgrade settlement.

11.2 Spatial Feature Attribute

Definition

IslandLine represents the linear footprint or alignment of an island or island segment within a river or backwater area when the feature is too narrow or detailed to be mapped effectively as a polygon at project scale. It captures the engineered or restored island crest or centerline used for analysis, design reference, or cartographic depiction.

Purpose

IslandLine features are used to represent long, narrow, or small island structures where polygon mapping is impractical, while still documenting their location, extent, and functional role in influencing flow, wave action, and habitat conditions. They provide a simplified linear representation that can be used in hydraulic analysis, design layouts, and monitoring of island performance over time.

Applicability

IslandLine is applicable where island features are present but are either too small to digitize as polygons at the project mapping scale or where a linear representation (e.g., crest line, centerline, or control alignment) is needed for modeling, design, or management. Typical uses include narrow barrier islands, training-island alignments, or thin island ridges within Upper Mississippi River System (UMRS) restoration projects. (UMRR Design Handbook, 2012)

Notes

Restoration features now have a Program_UID field to distinguish between NESP and UMRR features. Aliases without program identification will be created for any field that currently has UMRR in the name, so they can be used for NESP or UMRR data.

Attribute criteria for each field includes the following:

- Attribute Status – defines whether the field requires a value or if it is optional based on available information.

- Phases – indicates the earliest time in the project lifecycle values for this attribute are available.
- Digitizing requirements – indicates when the field value should be entered in the attribute table.

Geometry Note: No Z values are needed for this feature.

IslandLine

(line) – Linear representation of island structures engineered with a variety of techniques and materials that serve as barriers to flow, wind, and waves.

11.3 Feature Attribute Criteria

#

Attributes

Attribute Status

Phase

Digitizing Requirement

1

Program_UID

Required

Feasibility

At feature creation

2

PROJECT_ID

Required

Feasibility

At feature creation

3

Type

Required

Feasibility

At feature creation

4

Subtype

Required

Feasibility

At feature creation

5

RestorFeatStatus

Required

Design

At feature creation

6

sdsFeatureName

Optional

Design

As required and available

7

sdsFeatureDescription

Optional

Design

As required and available

8

RestorFeatVdatum

Optional

Design

As required and available

9

NumberPumps

Optional

Design

As required and available

10

PumpConfiguration

Optional

Construction

As required and available

11

PumpDrive

Optional

Construction

As required and available

12

PumpType

Optional

Construction

As required and available

13

MaxDesignHeadFT

Optional

Construction

As required and available

14

PumpCapacityCFS

Optional

Construction

As required and available

15

PumpElevation

Optional

Construction

As required and available

16

InvertElevation

Optional

Construction

As required and available

17

PumpAgeYRS

Optional

Construction

As required and available

18

DateBuilt

Optional

Construction

As required and available

19

Station

Optional

Construction

As required and available

20

sdsMetaDataID

Optional

Construction

As required and available

21

RestorFeatIDPK

Optional

Construction

As required and available

22

sdsID

Optional

Construction

As required and available

23

mediaIDFK

Optional

Construction

As required and available

11.4 Parameters

Program_UID

Designate program NESP or UMRR.

Project_ID (alias: PROJECT_ID)

Foreign key to the unique NESP/UMRR project ID.

Type (alias)

A broad classification used to group NESP/UMRR restoration features into categories of similar function.

> **Note:** The Type/Subtype section below appears to have been copied from PumpStation.
> For IslandLine, you will likely want to change these to `Island Creation` and appropriate island subtypes to match the polygon Island feature class. They are shown here as provided.

Value

Description

Water Level Management

Measures taken to imitate natural low flow river conditions with the goal of improving wetland habitats.

SubType (alias: Subtype)

A classification used to group NESP/UMRR restoration features into specific design feature categories.

Value

Description

Pump Station

Water pump infrastructure, comprised of a pad, pump, and pump enclosure, used to introduce or remove water from a MSMU or backwater lake.

RestorFeatStatus

A description of the status of the individual restoration. It also includes the party responsible for the individual feature.

Value

Description

USACE Planned

Feature is in a USACE planning stage.

USACE Built

Feature has been built by USACE. If a feature has been planned and built by USACE, once constructed, it will always be considered USACE built even if it has been repaired under later project stages.

USACE Enhanced

The function of an existing feature that was not built by USACE has been enhanced by a feature that was USACE built or repaired.

USACE Repaired

An existing feature that was not built by USACE has been directly improved or repaired.

Other Planned

A feature that is planned by another organization or government agency.

Other Built

Feature has been built by another organization or government agency and is a integral part of the UMRR project.

Other Enhanced

The function of an existing feature, that was built by another organization or government agency, has been enhanced by a feature that was built or repaired by a non-USACE organization or government agency.

Other Repaired

An existing feature that has been directly improved or repaired by another organization or government agency.

sdsFeatureName

The common name of the feature.

sdsFeatureDescription

A narrative describing the feature.

RestorFeatVdatum

Vertical datum of elevations recorded in the attribute table or used as feature z values.

Value

Description

MSL 1912

Mean Sea Level of 1912.

NGVD 1929

National Geodetic Vertical Datum of 1929.

NAVD 1988

North American Vertical Datum of 1988.

Other

Other, note datum in sdsFeatureDescription.

Unknown

Unknown vertical datum.

11.5 Pump-related Attributes (as currently defined)

> These attributes are included because they appear in your provided IslandLine schema, but they are typically associated with PumpStation. If IslandLine is not intended to store pump information, these may be removed or repurposed in your data model.

NumberPumps

Number of a specific type of pumps in the pump station.

PumpConfiguration

Pump configuration generally referring to a horizontally or vertically oriented pump. Area available for pump placement usually dictates this.

Value

Description

Horizontal

Horizontal pump configuration.

Vertical Line Shaft

Vertical line shaft pump configuration.

Vertical Submersible

Vertical submersible pump configuration.

Unknown

Unknown pump configuration.

NA

Not applicable.

PumpDrive

The method used to drive the pump action.

Value

Description

Diesel engine

Diesel engine pump drive.

Electric motor

Electric motor pump drive.

Natural gas engine

Natural gas engine pump drive.

Unknown

Unknown.

NA

Not applicable.

PumpType

The common pump type.

Value

Description

Axial flow

Axial flow pump type.

Centrifugal

Centrifugal pump type.

Mixed flow

Mixed flow pump type.

Unknown

Unknown.

NA

Not applicable.

MaxDesignHeadFT

The water elevation difference corresponding to the maximum design head of the pump (feet).

PumpCapacityCFS

Pumping capacity in cubic feet per second (if multiple pumps, this should be the total capacity).

PumpElevation

The midpoint elevation of the pump impeller or propeller.

InvertElevation

The lowest sill elevation (e.g., intake or discharge invert).

PumpAgeYRS

Age of the pump in years.

DateBuilt

Date the pump station (or installation) was built.

Station

Stationing of the feature, typically referenced to a project baseline or alignment (e.g., 10+00).

Horsepower

Horsepower of the pump.

FlowRate

Flow rate of the pump (GPM).

PumpStationType

Type of pump station installation.

Value

Description

Permanent Pump Station

Permanent pump station.

Portable Pump

Portable pump station.

sdsMetaDataID

References attached metadata.

RestorFeatIDPK

Primary key used to uniquely identify the restoration feature record.

sdsID

Identifier used to link the feature to supporting SDS (spatial data system) records or related datasets.

mediaIDFK

Foreign key used to reference associated media (e.g., photographs, design drawings, as-built plans).

12 Island

12.1 Spatial & Geometric Definition

- Geometry Type: Polygon (Restoration_Island_Polygon).
- Spatial Delineation: Subaerial landmass polygon constructed above normal flat pool within broad, shallow river expanses.
- Architectural Typology: Barrier islands, multi-island archipelagos, horseshoe bays, and interior spits.

Primary Restoration & Engineering Objectives

- Wind-Wave Fetch Reduction: Break wind-wave generation across shallow open bays, reducing bottom orbital wave resuspension and restoring water clarity for submersed aquatic vegetation (SAV).
- Terrestrial Habitat Creation: Provide secure, predator-isolated nesting habitat for migratory waterfowl, shorebirds, turtles, and bottomland hardwood trees.
- Sediment Trapping: Create sheltered leeward lagoons that promote fine sediment drop-out and organic soil building.

Typical Cross-Sections, Dimensions & Sizing Criteria

- Crest Elevations: Tiered multi-level profiles:
- High Ridges (5 to 10 ft above pool): Designed above the 10- to 25-year flood stage to preserve terrestrial flood refugia and bottomland hardwood mast forests.
- Low Sills / Wetland Benches (1 to 3 ft above pool): Frequently overtopped to sustain emergent marsh fringes.
- Base Widths & Footprints: Base widths of 100 to 400 ft; total individual footprints ranging from 2 to 50+ acres.
- Side Slopes: Windward wave-impact faces sloped at 1V:3H to 1V:4H; leeward slopes shaped gently at 1V:8H to 1V:20H down to shallow littoral shelves.

Construction & Implementation Methodology

- Material Sourcing & Sequencing: Fine to medium granular dredged sand mechanically or hydraulically placed to form the structural core.
- Cohesive Topsoil Caps: Capping the sand core with a 1.0- to 2.0-ft layer of fine-grained cohesive dredge silt/clay to support plant rooting and resist wind deflation.
- Windward Revetment Placement: Placing non-woven geotextile and Class II–III riprap along exposed upstream and windward shorelines concurrently with fill operations.

Quality Control, Inspection & Long-Term Maintenance

- Topographic Lidar & RTK Surveys: Post-construction as-built terrain modeling to verify settled elevations against contract cut/fill balances.
- Shoreline Stability Audits: Biannual visual inspections to detect riprap displacement, wave notch cutting, or flank erosion following major spring river floods.

12.2 Spatial Feature Attributes

Definition

Island features are engineered or restored landforms within a river or backwater area, constructed or modified using a variety of techniques and materials to provide physical barriers to flow, wind, and waves, and to support habitat and geomorphic functions.

Purpose

The loss of islands in the Upper Mississippi River is primarily due to human-induced alterations to the river's natural hydrology and sediment dynamics. Dams, channelization, sediment loss, and altered floodplain connectivity have disrupted the processes that create and maintain islands. As a result, islands are increasingly vulnerable to erosion, lack of sediment deposition, and other environmental stresses, leading to their gradual disappearance. This loss impacts biodiversity and the overall health of the river ecosystem, emphasizing the need for restoration and management efforts to protect and rebuild these important habitats. Island creation serves to restore structural diversity, reduce wave and current energy, improve sediment processes, and provide critical habitat for fish, wildlife, and plants.

Applicability

Island creation serves to restore and enhance river ecosystems, providing diverse habitats for wildlife, improving sediment and water quality processes, supporting floodplain and aquatic species health, and promoting biodiversity. Islands can be incorporated into restoration designs to protect shorelines, create sheltered backwaters, influence flow paths, and support vegetation and nesting habitat. (UMRR Design Handbook, 2012)

Notes

Restoration features now have a Program_UID field to distinguish between NESP and UMRR features. Aliases without program identification will be created for any field that currently has UMRR in the name, so they can be used for NESP or UMRR data.

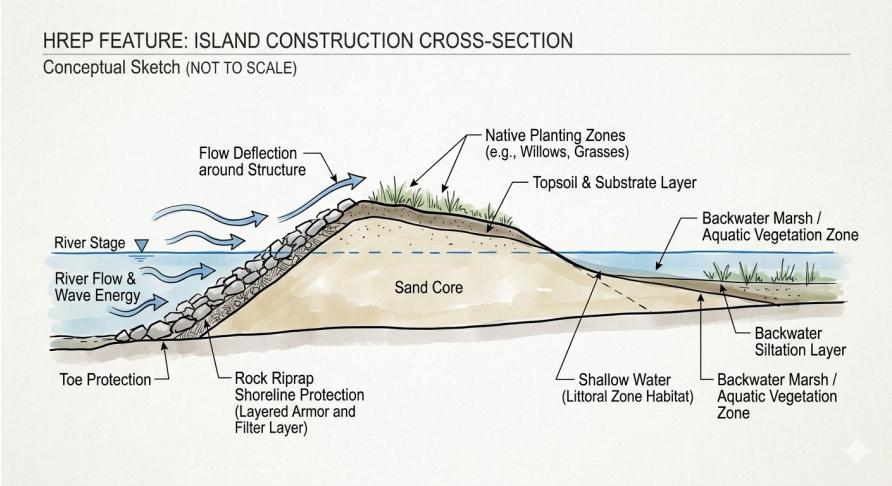
Attribute criteria for each field includes the following:

- Attribute Status – defines whether the field requires a value or if it is optional based on available information.
- Phases – indicates the earliest time in the project lifecycle values for this attribute are available.
- Digitizing requirements – indicates when the field value should be entered in the attribute table.

Geometry Note: Polygon nodes should have z values; elevations should be taken from as-built survey or design documents.

Island

(polygon) – Island structures engineered with a variety of techniques and materials that serve as barriers to flow, wind, and waves.



12.3 Feature Attribute Criteria

#

Attributes

Attribute Status

Phase

Digitizing Requirement

1

Program_UID

Required

Feasibility

At feature creation

2

PROJECT_ID

Required

Feasibility

At feature creation

3

Type

Required

Feasibility

At feature creation

4

Subtype

Required

Feasibility

At feature creation

5

RestorFeatStatus

Required

Design

At feature creation

6

sdsFeatureName

Optional

Design

As required and available

7

sdsFeatureDescription

Optional

Design

As required and available

8

RestorFeatVdatum

Optional

Design

As required and available

9

WaterOvertopping

Optional

Design

As required and available

10

PrimaryMaterial

Optional

Construction

As required and available

11

SecondaryMaterial

Optional

Construction

As required and available

12

DredgeMaterialPlacement

Optional

Construction

As required and available

13

Acres

Optional

Construction

As required and available

14

sdsMetaDataID

Optional

Construction

As required and available

15

RestorFeatIDPK

Optional

Construction

As required and available

16

sdsID

Optional

Construction

As required and available

17

mediaIDFK

Optional

Construction

As required and available

12.4 Parameters

Program_UID

Designate program NESP or UMRR.

Project_ID (alias: PROJECT_ID)

Foreign key to the unique NESP/UMRR project ID.

Type (alias)

A broad classification used to group NESP/UMRR restoration features into categories of similar function.

Value

Description

Island Creation

Measures taken to construct islands in rivers.

SubType (alias: Subtype)

A classification used to group NESP/UMRR restoration features into specific island design feature categories.

Value

Description

Barrier island

Island structures engineered with a variety of techniques and materials that serve as barriers to flow, wind, and waves.

Seed island

Rock or log structures placed perpendicular to currents in areas where coarse sediment transport is occurring, used to create islands via sediment accumulation due to reduced velocities.

Nourished seed island

Rock or log structures perpendicular to currents with sand placed on the upstream or downstream side; currents reshape sand naturally, used where coarse sediment transport is low.

Rock sill

Short, steep ramp comprised of boulders created to circumvent in-channel barriers to natural fish movement.

Sand tip

Low structures overtopped more frequently than islands, used to increase flow into areas during floods or to maintain flood flow into secondary channels; may include geotechnical membrane in the upstream sill to reduce seepage.

Sand flat

Areas of sand along islands precisely graded to be seasonally inundated to very shallow depths (0.1–0.2 ft) to create microtopography and habitat diversity.

Loafing structure

Large woody debris anchored into rock structures or placed along shorelines to provide loafing habitat for waterfowl, turtles, and other fauna, as well as fish habitat and enhanced shoreline aesthetics.

Rock/Log island

Structures constructed using rock to anchor large woody debris in the river channel, providing habitat diversity and loafing structures while improving aesthetics.

RestorFeatStatus

A description of the status of the individual restoration. It also includes the party responsible for the individual feature.

Value

Description

USACE Planned

Feature is in a USACE planning stage.

USACE Built

Feature has been built by USACE. If a feature has been planned and built by USACE, once constructed, it will always be considered USACE built even if it has been repaired under later project stages.

USACE Enhanced

The function of an existing feature that was not built by USACE has been enhanced by a feature that was USACE built or repaired.

USACE Repaired

An existing feature that was not built by USACE has been directly improved or repaired.

Other Planned

A feature that is planned by another organization or government agency.

Other Built

Feature has been built by another organization or government agency and is an integral part of the UMRR project.

Other Enhanced

The function of an existing feature, built by another organization or government agency, has been enhanced by a feature that was built or repaired by a non-USACE organization or government agency.

Other Repaired

An existing feature that has been directly improved or repaired by another organization or government agency.

sdsFeatureName

The common name of the feature.

sdsFeatureDescription

A narrative describing the feature.

RestorFeatVdatum

Vertical datum of elevations recorded in the attribute table or used as feature z values.

Value

Description

MSL 1912

Mean Sea Level of 1912.

NGVD 1929

National Geodetic Vertical Datum of 1929.

NAVD 1988

North American Vertical Datum of 1988.

Other

Other; note datum in sdsFeatureDescription.

Unknown

Unknown vertical datum.

WaterOvertopping

Description of the relationship between the island feature and water level.

Value

Description

AlwaysDry

Not covered at high water under average meteorological conditions.

AlwaysSubmerged

Remains covered by water at all times under average meteorological conditions.

Awash

Flush with or washed by waves at low water under average meteorological conditions.

CoversAndUncovers

Projection from the bottom that periodically extends above and is submerged below the water surface.

Floating

Resting or moving on the surface of a water body.

PartlySubmergedAtHighWater

Partially covered and partially dry at high water.

SubjectToInundationOrFlooding

Area or structure periodically covered by flood waters.

Unknown

Relationship to water level is unknown.

NA

Not applicable.

PrimaryMaterial

Primary construction material used for the island feature.

Value

Description

Sand

Sand.

Rock

Rock.

Fines

Fine sediment including clay and clay mixtures.

Timber/Wood

Timber or wood, including logs or root wads.

Earthen

Mixture of sand and fines.

Concrete

Concrete.

SecondaryMaterial

Secondary construction material, if applicable.

Value

Description

Sand

Sand.

Rock

Rock.

Fines

Fine sediment including clay and clay mixtures.

Timber/Wood

Timber or wood, including logs or root wads.

Earthen

Mixture of sand and fines.

Concrete

Concrete.

DredgeMaterialPlacement

Describes whether dredged material was placed to create or augment the island feature.

Value

Description

Yes

Dredged material was placed to create or augment the feature.

No

No dredged material was used.

NA

Not applicable.

Unknown

Use of dredged material is unknown.

Acres

Acreage of the island feature.

sdsMetaDataID

References attached metadata.

RestorFeatIDPK

Primary key used to uniquely identify the restoration feature record.

sdsID

Identifier used to link the feature to supporting SDS (spatial data system) records or related datasets.

mediaIDFK

Foreign key used to reference associated media (e.g., photographs, design drawings, construction records, as-built plans) documenting the island feature.

13 Moist Soil Management Unit

13.1 Spatial & Geometric Definition

- Geometry Type: Polygon (MSU_Cell_Boundary).
- Spatial Delineation: Enclosed, impounded floodplain basin bounded by perimeter containment embankments, internal ditches, and water control structures.
- Ecological Classification: Shallow seasonally flooded palustrine emergent wetland complex.

Primary Restoration & Engineering Objectives

- Targeted Seed & Tuber Production: Provide precise, independent hydrologic drawdowns and reflooding to stimulate high-energy forage plants (smartweeds, wild millets, sedges, sprangletop).
- Migratory Waterfowl Refueling: Provide shallow foraging water depths (2.0 to 8.0 in) timed precisely to match spring and autumn waterfowl migration pulses.
- Shorebird Staging Flats: Expose extensive mudflats during late summer (July–August) for migrating northbound and southbound shorebirds.

13.2 Typical Cross-Sections, Dimensions & Sizing Criteria

- Cell Surface Areas: Individual management cells typically scale from 10 to 80 acres.
- Topographic Grading: Graded with gentle interior slopes (<0.1% grade) toward a central drainage swale network (bottomwidths10 to 20 ft) to ensure complete gravity drainage without water pocketing.
- Operating Depth Envelope: Maximum impounded water depth of 1.5 to 3.0 ft; drawdowns expose 100% of cell floor to atmospheric drying.

Construction & Implementation Methodology

- Precision Earthmoving: Laser-guided bulldozers and low-ground-pressure earthmovers shaping interior swales and cell bottom micro-topography.
- Clay Liner Compaction: Where underlying alluvium contains excessive sand lenses, interior floors are sealed with a compacted 1.0- to 2.0-ft cohesive clay blanket to prevent downward water loss.

Quality Control, Inspection & Long-Term Maintenance

- Annual Water Management Plans: Rigorous hydro-period cycling: slow spring drawdowns (March–May), summer soil drying and germination (June–August), and gradual fall re-flooding (September–November).
- Discing & Succession Setback: Rotational tractor discing or controlled burning every 3 to 5 years to reset plant succession and eliminate monotypic willow or cattail stands.

Spatial Feature Attributes ### Definition Areas or units designed to enhance wetland habitat through the use of water containment, water supply, and water control structures.

Purpose

The basic operating plan for a Moist Soil Management Unit (MSMU) is to keep water out in the late spring and summer and to gradually flood the area in the fall. In a multiple-cell system, independent control of water levels in each cell is preferred to allow targeted habitat management.

Applicability

Moist Soil Management Units are typically designed to include water containment, water supply, and water control structures. Water containment is provided by construction of exterior berms, interior berms, and overflow spillways, which are used to impound water during seasonal waterfowl migrations or to keep water out of the impounded area. Water supply may be provided by either river water or groundwater through the use of a pump station or well, respectively. Water control structures are utilized to maintain desired water elevations throughout the year. (UMRR Design Handbook, 2012)

Notes

Restoration features now have a Program_UID field to distinguish between NESP and UMRR features. Aliases without program identification will be created for any field that currently has UMRR in the name, so they can be used for NESP or UMRR data.

Attribute criteria for each field includes the following:

- Attribute Status – defines whether the field requires a value or if it is optional based on available information.
- Phases – indicates the earliest time in the project lifecycle values for this attribute are available.
- Digitizing requirements – indicates when the field value should be entered in the attribute table.

Geometry Note: No Z values are needed for this feature.

MoistSoilMgmtUnit

(polygon) – Area or area subdivision that is managed to create moist soil units.

13.3 Feature Attribute Criteria

#

Attribute

Attribute Status

Phase

Digitizing Requirements

1

Program_UID

Required

Feasibility

At feature creation

2

PROJECT_ID (Alias)

Required

Feasibility

At feature creation

3

Type (Alias)

Required

Feasibility

At feature creation

4

Subtype (Alias)

Required

Feasibility

At feature creation

5

RestorFeatStatus

Required

Feasibility

At feature creation

6

sdsFeatureName

Optional

Design

As required and available

7

sdsFeatureDescription

Optional

Design

As required and available

8

RestorFeatVdatum

Optional

Design

As required and available

9

Acres

Optional

Design

As required and available

10

Depth

Optional

Design

As required and available

11

WaterControlType

Optional

Design

As required and available

12

WaterControlMethod

Optional

Design

As required and available

13

sdsMetaDataID

Optional

Construction

As required and available

14

RestorFeatIDPK

Optional

Construction

As required and available

15

sdsID

Optional

Construction

As required and available

16

mediaIDFK

Optional

Construction

As required and available

13.4 Parameters

Program_UID

Designate program NESP or UMRR.

Project_ID (alias: PROJECT_ID)

Foreign key to the unique NESP/UMRR project ID.

Type (alias)

A broad classification used to group NESP/UMRR restoration features into categories of similar function.

Value

Description

Water Level Management

Measures taken to manage and manipulate water levels with the goal of improving wetland habitats.

SubType (alias: Subtype)

A classification used to group NESP/UMRR restoration features into specific design feature categories.

Value

Description

Moist soil management unit

Units designed to enhance wetland habitat through use of water containment, water supply, and water control structures.

Backwater lake

Units designed to enhance aquatic habitat through use of water containment, water supply, and water control structures.

Cell

Sub-unit of a moist soil unit, bounded by exterior and/or interior berms, allowing targeted water level management within a MSMU.

RestorFeatStatus

A description of the status of the individual restoration. It also includes the party responsible for the individual feature.

Value

Description

USACE Planned

Feature is in a USACE planning stage.

USACE Built

Feature has been built by USACE. If a feature has been planned and built by USACE, once constructed, it will always be considered USACE built even if repaired under later project stages.

USACE Enhanced

The function of an existing feature not built by USACE has been enhanced by a feature that was USACE built or repaired.

USACE Repaired

An existing feature not built by USACE has been directly improved or repaired.

Other Planned

A feature that is planned by another organization or government agency.

Other Built

Feature has been built by another organization or government agency and is an integral part of the UMRR project.

Other Enhanced

The function of an existing feature, built by another organization or government agency, has been enhanced by a feature that was built or repaired by a non-USACE organization or government agency.

Other Repaired

An existing feature that has been directly improved or repaired by another organization or government agency.

sdsFeatureName

The common name of the feature.

sdsFeatureDescription

A narrative describing the feature.

RestorFeatVdatum

Vertical datum of elevations recorded in the attribute table or used as feature z values.

Value

Description

MSL 1912

Mean Sea Level of 1912.

NGVD 1929

National Geodetic Vertical Datum of 1929.

NAVD 1988

North American Vertical Datum of 1988.

Other

Other; note datum in sdsFeatureDescription.

Unknown

Unknown vertical datum.

Acres

Acreage of the feature.

Depth

Typical or design depth of the feature (e.g., target water depth), recorded in appropriate units (commonly feet).

WaterControlType

General classification of how water control is achieved.

Value

Description

Passive

Water control depends on natural processes (e.g., seasonal inundation) or passive engineering structures (e.g., spillway or notch).

Active

Water control depends on human actions (e.g., pump station operation, stoplog structures).

WaterControlMethod

Specific water control technique used.

Value

Description

Overflow spillway

Section of embankment engineered to provide controlled overtopping of an embankment system, typically at the downstream end of a site at an elevation lower than the exterior berm.

Pump

Water pump infrastructure (pad, pump, enclosure) used to introduce or remove water from a MSMU or backwater lake.

Well

Vertical hole drilled into the ground to access or influence groundwater movement, used to introduce or remove water from a MSMU or backwater lake.

Stoplog

Water management structure that allows control of water level or flow rate into or out of MSMUs and backwater lakes by adding or removing stoplogs.

Other

Other method; specify in sdsFeatureDescription.

sdsMetaDataID

References attached metadata.

RestorFeatIDPK

Primary key used to uniquely identify the restoration feature record.

sdsID

Identifier used to link the feature to supporting SDS (spatial data system) records or related datasets.

mediaIDFK

Foreign key used to reference associated media (e.g., photographs, design drawings, operating plans, as-built plans) documenting the moist soil management unit.

14 Notch

14.1 Spatial & Geometric Definition

- Geometry Type: Polygon or Point (Notch_Geometry).
- Spatial Delineation: Geometric cut, depression, or gap engineered into rock training structures, wing dams, closing dams, or earthen levee crests.
- Cross-Sectional Typology: Trapezoidal, parabolic, or rectangular weir cuts.

Primary Restoration & Engineering Objectives

- Secondary Channel Revitalization: Pass controlled baseflow through closing dams to refresh side channels without destabilizing main-channel navigation drafts.
- Bathymetric Scour Generation: Concentrate localized flow vectors through the notch throat, carving deep downstream plunge pools (10 to 25 ft deep) that function as prime overwintering fish holes.
- Sediment Purging: Re-energize sediment transport capacity through backwaters to purge deposited silt plugs.

Typical Cross-Sections, Dimensions & Sizing Criteria

- Throat Widths: Bottom widths typically range from 10 to 50 ft; side slopes cut at 1V:2H to 1V:3H.
- Depth of Invert: Invert set 1.0 to 4.0 ft below the parent weir crest or normal flat pool elevation.
- Hydraulic Conveyance: Sized using broad-crested weir formulas (Q=C d LH 3/2) to restrict diverted river discharge to calibrated project thresholds (typically <5% of total channel discharge).

Construction & Implementation Methodology

- Marine Excavation: Existing stone removed using barge-mounted hydraulic excavators equipped with rock buckets or heavy clamshells.
- Bed & Abutment Armoring: Installing heavy riprap transitions along notch edges and lining the downstream plunge impact zone with Class IV–V stone to halt headcut unraveling.

Quality Control, Inspection & Long-Term Maintenance

- Single-Beam Sonar Soundings: Annual hydrographic cross-sections through the notch and downstream scour basin to verify that scour hole depth remains stable and does not undermine the main weir abutments.

14.2 Spatial Feature Attribute

Definition

A notch is a deliberate opening cut into a riverine flow structure or levee (typically a dike) used to control the flow of water into and out of a low-flow area or adjacent floodplain.

Purpose

A notch creates a controlled gap or opening in the structure. Its purpose is to regulate water flow, manage river levels, and prevent damage from excessive water buildup. Notching can support flow regulation, sediment management, habitat creation, and flood control by selectively passing water and sediment and reconnecting channels and floodplains.

Applicability

Notches are applicable where targeted hydraulic connections are needed between the main channel and backwaters, side channels, or floodplain areas. They are commonly used to improve habitat, reduce sedimentation, restore natural inundation patterns, and modify river training structures or levees to better meet environmental objectives within Upper Mississippi River System (UMRS) restoration projects. (UMRR Design Handbook, 2012)

Notes

Restoration features now have a Program_UID field to distinguish between NESP and UMRR features. Aliases without program identification will be created for any field that currently has UMRR in the name, so they can be used for NESP or UMRR data.

Attribute criteria for each field includes the following:

- Attribute Status – defines whether the field requires a value or if it is optional based on available information.
- Phases – indicates the earliest time in the project lifecycle values for this attribute are available.
- Digitizing requirements – indicates when the field value should be entered in the attribute table.

Geometry Note: No Z values are needed for this feature.

Notch

(point) – A notch in a riverine flow structure or levee used to control the flow of water into and out of the low-flow area.

14.3 Feature Attribute Criteria

#

Attribute

Attribute Status

Phase

Digitizing Requirements

1

Program_UID

Required

Feasibility

At feature creation

2

PROJECT_ID (Alias)

Required

Feasibility

At feature creation

3

Type (Alias)

Required

Feasibility

At feature creation

4

Subtype (Alias)

Required

Feasibility

At feature creation

5

RestorFeatStatus

Required

Feasibility

At feature creation

6

sdsFeatureName

Optional

Design

As required and available

7

sdsFeatureDescription

Optional

Design

As required and available

8

RestorFeatVdatum

Optional

Design

As required and available

9

ConstructionDate

Optional

Design

As required and available

10

NotchElevation

Optional

Design

As required and available

11

NotchWidthFT

Optional

Design

As required and available

12

Station

Optional

Construction

As required and available

13

sdsMetaDataID

Optional

Construction

As required and available

14

RestorFeatIDPK

Optional

Construction

As required and available

15

sdsID

Optional

Construction

As required and available

16

mediaIDFK

Optional

Construction

As required and available

14.4 Parameters

Program_UID

Designate program NESP or UMRR.

Project_ID (alias: PROJECT_ID)

Foreign key to the unique NESP/UMRR project ID.

Type (alias)

A broad classification used to group NESP/UMRR restoration features into categories of similar function.

Value

Description

Channel Modifications

Measures taken to train the main channel of a river and to modify secondary channels.

Floodplain Restoration

Measures taken to improve natural floodplain function and habitat.

SubType (alias: Subtype)

A classification used to group NESP/UMRR restoration features into specific notch design feature categories.

Value

Description

Wing dam notch

Wing dam modification that significantly lowers a section of the dam to support diverse habitats while maintaining navigation channel function.

Closure structure notch

Closure structure modification that significantly lowers a section of the structure, allowing decreased sedimentation and enhanced habitat.

Dike notch

Dike modification that significantly lowers a section of the structure, allowing decreased sedimentation and enhanced habitat.

Levee notch

A low spot excavated in a frontline levee to allow inundation of the area behind the levee above a specific flow.

RestorFeatStatus

A description of the status of the individual restoration. It also includes the party responsible for the individual feature.

Value

Description

USACE Planned

Feature is in a USACE planning stage.

USACE Built

Feature has been built by USACE. If a feature has been planned and built by USACE, once constructed, it will always be considered USACE built even if repaired under later project stages.

USACE Enhanced

The function of an existing feature not built by USACE has been enhanced by a USACE-built or repaired feature.

USACE Repaired

An existing feature not built by USACE has been directly improved or repaired.

Other Planned

Feature planned by another organization or government agency.

Other Built

Feature built by another organization or government agency and integral to the UMRR project.

Other Enhanced

Function of a feature built by another organization has been enhanced by a feature built or repaired by a non-USACE organization or government agency.

Other Repaired

Existing feature directly improved or repaired by another organization or government agency.

sdsFeatureName

The common name of the feature.

sdsFeatureDescription

A narrative describing the feature.

RestorFeatVdatum

Vertical datum of elevations recorded in the attribute table or used as feature z values.

Value

Description

MSL 1912

Mean Sea Level of 1912.

NGVD 1929

National Geodetic Vertical Datum of 1929.

NAVD 1988

North American Vertical Datum of 1988.

Other

Other; note datum in sdsFeatureDescription.

Unknown

Unknown vertical datum.

ConstructionDate

The date of construction (or reconstruction) of the notch.

NotchElevation

Elevation of the bottom of the notch, referenced to the vertical datum specified in RestorFeatVdatum.

NotchWidthFT

Width of the notch in feet.

Station

Stationing of the feature, typically referenced to a project baseline or alignment (e.g., 10+00).

sdsMetaDataID

References attached metadata.

RestorFeatIDPK

Primary key used to uniquely identify the restoration feature record.

sdsID

Identifier used to link the feature to supporting SDS (spatial data system) records or related datasets.

mediaIDFK

Foreign key used to reference associated media (e.g., design drawings, construction photos, as-built plans) documenting the notch.

15 Pump Station

15.1 Spatial & Geometric Definition

- Geometry Type: Point or Polygon (Pump_Station_Structure).
- Spatial Delineation: Footprint encompassing cast-in-place concrete wet-wells, intake bays, discharge manifolds, electrical control buildings, and fuel/power infrastructure.
- Operational Classification: High-capacity, low-head axial-flow pumping systems.

Primary Restoration & Engineering Objectives

- Hydrologic Independence: Allow rapid filling or complete dewatering of moist soil units and managed wetland cells regardless of ambient river flood or drought stages.
- Dewatering Reliability: Drain impounded areas during spring flood stages when external river levels sit higher than the internal wetland ground elevation.
- Rapid Flood Protection: Evacuate internal runoff following heavy storm events to prevent premature drown-out of maturing wetland seed crops.

Typical Mechanical & Structural Engineering Specifications

- Pumping Capacity: Total station flow capacities range between 2,000 and 25,000 gpm (5 to 55 cfs), operating against total dynamic heads (TDH) of 6 to 18 ft.
- Pump Mechanics: Vertical axial-flow or mixed-flow electric submersible or lineshaft pumps. Sized with variable frequency drives (VFD) for adjustable flow rates.
- Reversible Manifold Piping: Multi-valve directional piping manifolds allowing a single pump train to draw water from the river to fill the cell, or draw from the cell to discharge out into the river.

Construction & Implementation Methodology

- Deep Foundation Systems: Reinforced concrete substructure founded on steel H-piling or driven ductile iron piles keyed into underlying load-bearing river sands to resist uplift hydrostatic pressures.
- Cofferdam Dewatering: Wet-well construction executed within sheetpile cofferdams dewatered by deep-well point systems.

Quality Control, Inspection & Long-Term Maintenance

- Intake Screening: Trash racks equipped with automated mechanical rakes or heavy bar screens (2- to 3-in clear spacing) to prevent log and debris ingestion.
- Winterization: Annual late-autumn dewatering of pump bowls, lubrication checks, and cycling of emergency backup diesel generators under full load.

15.2 Spatial Feature Attribute

Definition

A pump station is a facility that pumps water between two storage areas, a storage area and a river reach, or between different river reaches. It can also be used to transfer water within the same 2D Flow Area or between different cells within the same 2D Flow Area.

Purpose

Pump stations are used to introduce or remove water from backwater lakes, MSMUs, or other project areas using surface water (typically from a river) or groundwater. They provide controlled water level management, allowing project operators to manipulate hydrologic conditions to support habitat objectives, flood risk management, and operational needs.

Applicability

Pumping stations are used for removal of interior drainage from areas protected by local flood protection works and for water supply. Pumping stations can and should have negligible adverse environmental impacts. They can also be used to connect pumps, lift flows from lower to higher elevations, increase pressure in gravity flow distribution systems, and improve the selection and operation of water level management facilities within restoration projects. (UMRR Design Handbook, 2012)

Notes

Restoration features now have a Program_UID field to distinguish between NESP and UMRR features. Aliases without program identification will be created for any field that currently has UMRR in the name, so they can be used for NESP or UMRR data.

Attribute criteria for each field includes the following:

- Attribute Status – defines whether the field requires a value or if it is optional based on available information.
- Phases – indicates the earliest time in the project lifecycle values for this attribute are available.
- Digitizing requirements – indicates when the field value should be entered in the attribute table.

Geometry Note: No Z values are needed for this feature.

PumpStation

(point) – The location of a pump for the purpose of pumping water from the interior of a levee or embankment.

15.3 Feature Attribute Criteria

#

Attributes

Attribute Status

Phase

Digitizing Requirement

1

Program_UID

Required

Feasibility

At feature creation

2

PROJECT_ID

Required

Feasibility

At feature creation

3

Type

Required

Feasibility

At feature creation

4

Subtype

Required

Feasibility

At feature creation

5

RestorFeatStatus

Required

Design

At feature creation

6

sdsFeatureName

Optional

Design

As required and available

7

sdsFeatureDescription

Optional

Design

As required and available

8

RestorFeatVdatum

Optional

Design

As required and available

9

NumberPumps

Optional

Design

As required and available

10

PumpConfiguration

Optional

Construction

As required and available

11

PumpDrive

Optional

Construction

As required and available

12

PumpType

Optional

Construction

As required and available

13

MaxDesignHeadFT

Optional

Construction

As required and available

14

PumpCapacityCFS

Optional

Construction

As required and available

15

PumpElevation

Optional

Construction

As required and available

16

InvertElevation

Optional

Construction

As required and available

17

PumpAgeYRS

Optional

Construction

As required and available

18

DateBuilt

Optional

Construction

As required and available

19

Station

Optional

Construction

As required and available

20

sdsMetaDataID

Optional

Construction

As required and available

21

RestorFeatIDPK

Optional

Construction

As required and available

22

sdsID

Optional

Construction

As required and available

23

mediaIDFK

Optional

Construction

As required and available

15.4 Parameters

Program_UID

Designate program NESP or UMRR.

Project_ID (alias: PROJECT_ID)

Foreign key to the unique NESP/UMRR project ID.

Type (alias)

A broad classification used to group NESP/UMRR restoration features into categories of similar function.

Value

Description

Water Level Management

Measures taken to imitate natural low flow river conditions with the goal of improving wetland habitats.

SubType (alias: Subtype)

A classification used to group NESP/UMRR restoration features into specific design feature categories.

Value

Description

Pump Station

Water pump infrastructure, which may include a pad, pump(s), and pump enclosure, used to introduce or remove water from a MSMU or backwater lake.

RestorFeatStatus

A description of the status of the individual restoration. It also includes the party responsible for the individual feature.

Value

Description

USACE Planned

Feature is in a USACE planning stage.

USACE Built

Feature has been built by USACE. If a feature has been planned and built by USACE, once constructed, it will always be considered USACE built even if it has been repaired under later project stages.

USACE Enhanced

The function of an existing feature that was not built by USACE has been enhanced by a feature that was USACE built or repaired.

USACE Repaired

An existing feature that was not built by USACE has been directly improved or repaired.

Other Planned

A feature that is planned by another organization or government agency.

Other Built

Feature has been built by another organization or government agency and is an integral part of the UMRR project.

Other Enhanced

The function of an existing feature, that was built by another organization or government agency, has been enhanced by a feature that was built or repaired by a non-USACE organization or government agency.

Other Repaired

An existing feature that has been directly improved or repaired by another organization or government agency.

sdsFeatureName

The common name of the feature.

sdsFeatureDescription

A narrative describing the feature.

RestorFeatVdatum

Vertical datum of elevations recorded in the attribute table or used as feature z values.

Value

Description

MSL 1912

Mean Sea Level of 1912.

NGVD 1929

National Geodetic Vertical Datum of 1929.

NAVD 1988

North American Vertical Datum of 1988.

Other

Other; note datum in sdsFeatureDescription.

Unknown

Unknown vertical datum.

NumberPumps

Number of pumps of a specific type in the pump station.

PumpConfiguration

Pump configuration, generally referring to a horizontally or vertically oriented pump. The area available for pump placement usually dictates configuration.

Value

Description

Horizontal

Horizontal pump configuration.

Vertical Line Shaft

Vertical line shaft pump configuration.

Vertical Submersible

Vertical submersible pump configuration.

Unknown

Unknown pump configuration.

NA

Not applicable.

PumpDrive

The method used to drive the pump action.

Value

Description

Diesel engine

Diesel engine pump drive.

Electric motor

Electric motor pump drive.

Natural gas engine

Natural gas engine pump drive.

Unknown

Unknown pump drive.

NA

Not applicable.

PumpType

The common pump type.

Value

Description

Axial flow

Axial flow pump type.

Centrifugal

Centrifugal pump type.

Mixed flow

Mixed flow pump type.

Unknown

Unknown pump type.

NA

Not applicable.

MaxDesignHeadFT

The maximum design head of the pump, expressed as the water surface elevation difference (in feet) between suction and discharge at design conditions.

PumpCapacityCFS

Pumping capacity of the station in cubic feet per second (CFS). If multiple pumps are present, this should represent the total station capacity.

PumpElevation

Midpoint elevation of the pump impeller or propeller, referenced to RestorFeatVdatum.

InvertElevation

Lowest sill or invert elevation of the intake or discharge conduit, referenced to RestorFeatV-datum.

PumpAgeYRS

Age of the pump in years (may be derived from DateBuilt or last replacement date).

DateBuilt

Date the pump station (or pump installation) was built or substantially completed.

Station

Stationing of the feature, typically referenced to a project baseline or alignment (e.g., 10+00).

Horsepower

Horsepower rating of the pump or pump motor (if tracked as a separate attribute in implementation).

FlowRate

Flow rate of the pump in gallons per minute (GPM) for individual pump units, if applicable.

PumpStationType

Type of pump station installation.

Value

Description

Permanent Pump Station

Fixed, permanent pump station installation.

Portable Pump

Portable or temporary pump installation.

sdsMetaDataID

References attached metadata.

RestorFeatIDPK

Primary key used to uniquely identify the restoration feature record.

sdsID

Identifier used to link the feature to supporting SDS (spatial data system) records or related datasets.

mediaIDFK

Foreign key used to reference associated media (e.g., photographs, design drawings, pump curves, as-built plans) documenting the pump station.

16 Revetment

16.1 Spatial & Geometric Definition

- Geometry Type: Polygon or Polyline(Revetment_Armor_Band).
- Spatial Delineation: Armored protective blanket placed along littoral shoreline banks, island provar edges, wing dam abutments, and channel margins.
- Materials: Heavy quarried angular stone riprap, crushed aggregate bedding, and non-woven geotextile filter fabric.

Primary Restoration & Engineering Objectives

- Shoreline Erosion Arrest: Neutralize high hydraulic shear stresses (= RS) induced by commercial towboat propeller wash, vessel wakes, and flood currents.
- Wind-Wave Dissipation: Protect constructed earthen island fills and natural delta edges from wind-fetch wave impact and slumping.
- Littoral Micro-Cover: Stone voids provide interstitial refuge and feeding zones for macroinvertebrates and small fish species.

Typical Cross-Sections, Dimensions & Sizing Criteria

- Armor Sizing: USACE Class II to Class IV stone riprap (d 50 =12 to 18 in; top weight up to 1,500 lbs).
- Blanket Thickness: 1.5× to 2.0× the d 100 stone size (typically 18 to 30 in thick).
- Slope Geometry: Maximum stable slope angle is 1V:2H (1V:3H preferred).
- Toe Key-In: Armored toe trench excavated 3.0 to 5.0 ft into the channel bed, or a wide stone launching apron (5 to 10 ft wide) designed to self-launch and protect the bank toe against scour.

Construction & Implementation Methodology

- Marine Placement: Work conducted via spud barges using long-reach excavators equipped with hydraulic stone grapples or clamshell buckets to place stone smoothly without tearing geotextile.
- Filter Layer Installation: High-survivability non-woven geotextile placed under the stone to stop fine-grained subgrade soils from leaching out through riprap voids.

Quality Control, Inspection & Long-Term Maintenance

- Gradation Testing: Quarry riprap testing per ASTM standards to ensure durability, bulk specific gravity (2.60), and resistance to freeze-thaw cycles.
- Post-Flood Audits: Boat-based visual and sonar scans to detect riprap displacement, toe undermining, or ice plucking along the waterline.

16.2 Spatial Feature Attribute

Definition

A revetment is a cover or facing of erosion-resistant material placed directly on an existing slope, embankment, or dike. Unlike seawalls, revetments are not self-supporting and depend on the underlying soil for support.

Purpose

Revetments are used to stabilize river banks and protect shorelines from strong waves and currents. Bank revetments prevent erosion, maintain a stable channel, and support new habitat creation. (UMRR Design Handbook, 2012)

Applicability

The primary forces that affect shorelines are river currents and wind-driven wave action; ice action and waves created by towboats or recreational boats can also cause erosion. Revetment stabilization techniques can be employed singly or in combination to protect shoreline and add habitat diversity. Gradual side slopes and sand or mud soils can benefit turtles and waterbirds that nest, feed, and loaf on shorelines. Native plantings can be more aesthetically pleasing than traditional riprap, and traditional stabilization techniques can be refined (e.g., larger and